

Chapter 7

The Application Layer (应用层)



Contents

- DNS
- Electronic Mail (电子邮件)
- World Wide Web (万维网)
- Multimedia (多媒体) [*]
- Network Management [*]

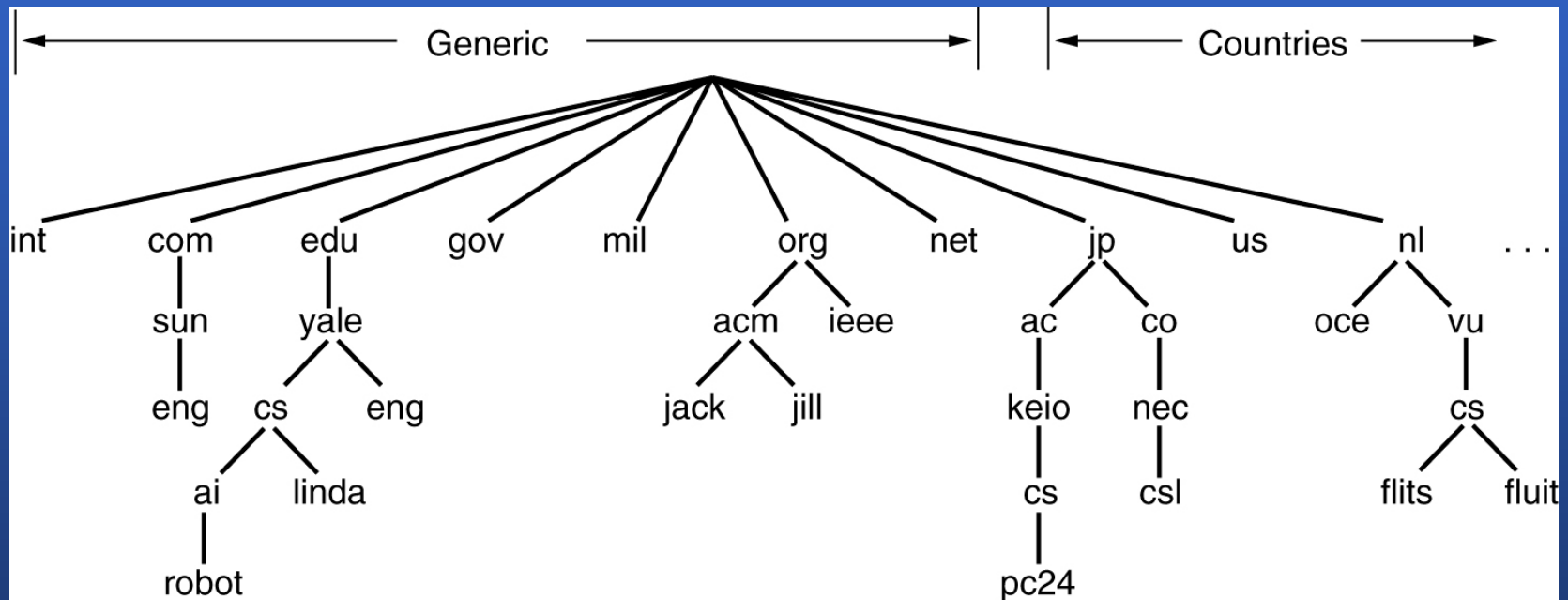


7.1 DNS – The Domain Name System

- The DNS Name Space
- Resource Records
- Name Servers



7.1.1 The DNS Name Space



A portion of the Internet domain name space.

The DNS Name Space(2)

- 顶层域分为两大类：一般的和国家的
 - 一般的：com, edu, gov, int, mil, net, org ...
 - 国家的：cn, ca, uk, jp, fr, ...
- 每个域由它往上到根的路径命名，成员由句点“.”分隔。 cs.zju.edu.cn
- 域名对大小写不敏感(**case insensitive**):
 - edu和EDU相同
- 成员名最长63个字符，路径名不能超过255个字符
- Domain names can be either **absolute or relative**



7.1.2 Resource Records

Type	Meaning	Value
SOA	Start of Authority	Parameters for this zone
A	IP address of a host	32-Bit integer
MX	Mail exchange	Priority, domain willing to accept e-mail
NS	Name Server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
HINFO	Host description	CPU and OS in ASCII
TXT	Text	Uninterpreted ASCII text

The principal DNS resource records types.

- DNS的实际功能就是把域名映射到资源记录上
- 资源记录的格式:
 - Domain_name, Time_to_live, Type, Class, Value
 - Type: 记录类型
 - 最重要的是A(地址)



Resource Records (2)

```
; Authoritative data for cs.vu.nl
cs.vu.nl.      86400  IN  SOA      star boss (952771,7200,7200,2419200,86400)
cs.vu.nl.      86400  IN  TXT      "Divisie Wiskunde en Informatica."
cs.vu.nl.      86400  IN  TXT      "Vrije Universiteit Amsterdam."
cs.vu.nl.      86400  IN  MX       1 zephyr.cs.vu.nl.
cs.vu.nl.      86400  IN  MX       2 top.cs.vu.nl.

flits.cs.vu.nl. 86400  IN  HINFO    Sun Unix
flits.cs.vu.nl. 86400  IN  A        130.37.16.112
flits.cs.vu.nl. 86400  IN  A        192.31.231.165
flits.cs.vu.nl. 86400  IN  MX       1 flits.cs.vu.nl.
flits.cs.vu.nl. 86400  IN  MX       2 zephyr.cs.vu.nl.
flits.cs.vu.nl. 86400  IN  MX       3 top.cs.vu.nl.
www.cs.vu.nl.   86400  IN  CNAME    star.cs.vu.nl
ftp.cs.vu.nl.   86400  IN  CNAME    zephyr.cs.vu.nl

rowboat         IN  A        130.37.56.201
                IN  MX       1 rowboat
                IN  MX       2 zephyr
                IN  HINFO    Sun Unix

little-sister   IN  A        130.37.62.23
                IN  HINFO    Mac MacOS

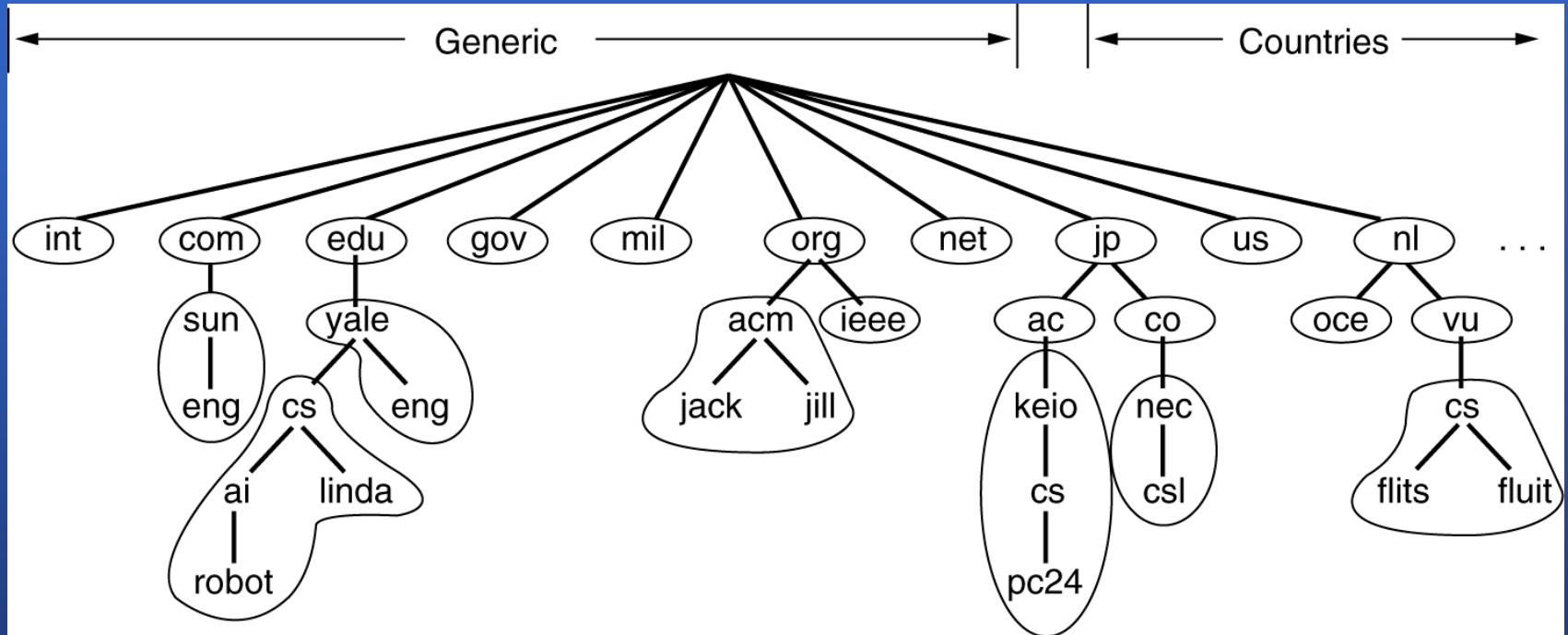
laserjet        IN  A        192.31.231.216
                IN  HINFO    "HP Laserjet IIISi" Proprietary
```

A portion of a possible DNS database for *cs.vu.nl*.





7.1.3 Name Servers



Part of the DNS name space showing the division into zones.

DNS名字空间被划分为一些不交叉的区域(zone)

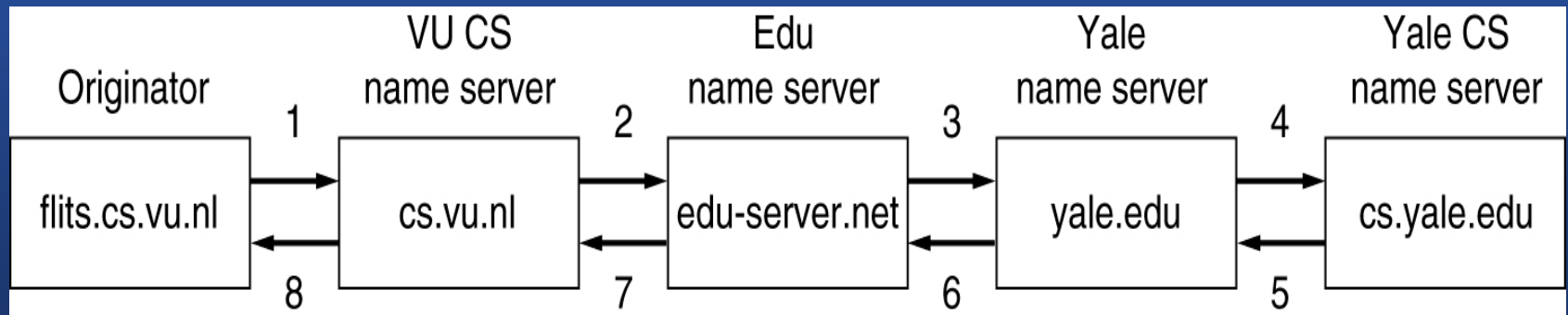
Name Servers (2)

- Authoritative record(权威记录): 记录管理机构获得的、总是正确的记录

- 缓存记录

- 远程名字的查询过程-[see fig 7-29]

- 递归查询



How a resolver looks up a remote name in eight steps

7.2 Electronic Mail

- Architecture and Services
- The User Agent
- Message Formats
- Message Transfer
- Final Delivery



Electronic Mail(2)

- 主要协议

- ARPANET: RFC 821 (传输协议) 和 RFC 822 (消息格式)
- CCITT X.400 → OSI: MOTIS 现 X.400 已基本消失



Electronic Mail (3)

Smiley	Meaning	Smiley	Meaning	Smiley	Meaning
: -)	I'm happy	= : -)	Abe Lincoln	: +)	Big nose
: -(	I'm sad/angry	=): -)	Uncle Sam	: -))	Double chin
: -	I'm apathetic	*<: -)	Santa Claus	: -{)	Mustache
; -)	I'm winking	<: -(	Dunce	#: -)	Matted hair
: -(O)	I'm yelling	(-:	Australian	8-)	Wears glasses
: -(*)	I'm vomiting	: -)X	Man with bowtie	C: -)	Large brain

Some smileys (emoticons, emotion icons).
They will not be on the final exam :-).

7.2.1 Architecture and Services

- 系统组成

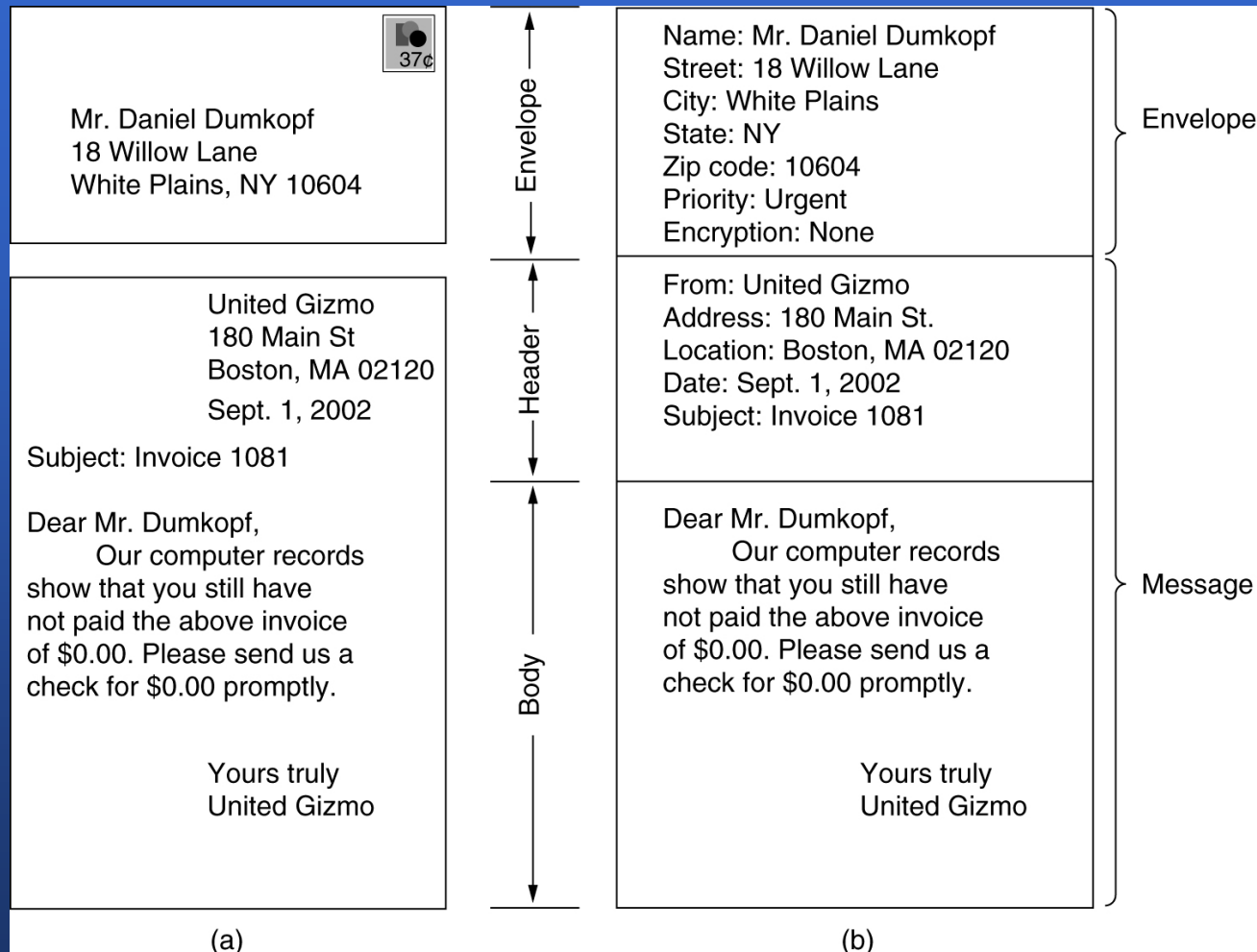
- 用户代理 和 消息传输代理 2个子系统

- Basic functions

- Composition(撰写)
 - Transfer(传输)
 - Reporting(报告)
 - Displaying(显示)
 - Disposition(处理)
 - 其它:转发、邮箱、收件人列表、挂号电子邮件、优先级、加密



7.2.2 The User Agent



Envelopes and messages. (a) Paper mail. (b) Electronic



Reading E-mail

#	Flags	Bytes	Sender	Subject
1	K	1030	asw	Changes to MINIX
2	KA	6348	trudy	Not all Trudys are nasty
3	K F	4519	Amy N. Wong	Request for information
4		1236	bal	Bioinformatics
5		104110	kaashoek	Material on peer-to-peer
6		1223	Frank	Re: Will you review a grant proposal
7		3110	guido	Our paper has been accepted
8		1204	dmr	Re: My student's visit

An example display of the contents of a mailbox.

- K – message is not new (be read), **k**ept in the mailbox
- A – message has already been **a**nswered

•F – message has been **f**orwarded to someone else



Mail Operations

- T – type message
- A – Answer message
- D – Delete message
- F – Forward message



7.2.3 Message Formats – RFC 822

- 基本ASCII格式 和 多媒体扩展格式
- RFC 822
 - 消息：基本信封+头字段+空行+主体
 - 与消息传输有关的RFC822头字段

Header	Meaning
To:	E-mail address(es) of primary recipient(s)
Cc:	E-mail address(es) of secondary recipient(s)
Bcc:	E-mail address(es) for blind carbon copies
From:	Person or people who created the message
Sender:	E-mail address of the actual sender
Received:	Line added by each transfer agent along the route
Return-Path:	Can be used to identify a path back to the sender

RFC 822 header fields related to message transport.



Message Formats – RFC 822 (2)

Header	Meaning
Date:	The date and time the message was sent
Reply-To:	E-mail address to which replies should be sent
Message-Id:	Unique number for referencing this message later
In-Reply-To:	Message-Id of the message to which this is a reply
References:	Other relevant Message-Ids
Keywords:	User-chosen keywords
Subject:	Short summary of the message for the one-line display

Some fields used in the RFC 822 message header.



7.2.4 MIME – Multipurpose Internet Mail Extensions

Problems with international languages:

- Languages with accents (French, German).
- Languages in non-Latin alphabets (Hebrew, Russian).
- Languages without alphabets (Chinese, Japanese).
- Messages not containing text at all (audio or images).



MIME (2)

- MIME - 多用因特网邮件扩展

- 基本思想：继续使用RFC 822的形式，但对消息主体增加结构，并为非ASCII消息定义编码结构
- 5种新的消息头部：

Header	Meaning
MIME-Version:	Identifies the MIME version
Content-Description:	Human-readable string telling what is in the message
Content-Id:	Unique identifier
Content-Transfer-Encoding:	How the body is wrapped for transmission
Content-Type:	Type and format of the content

RFC 822 headers added by MIME.



MIME (3)

- MIME — 多用因特网邮件扩展
 - 二进制消息编码方法:64个基本字符编码(24bit→4个6bit字符)
 - MIME类型和子类型-[see fig 7-12]
 - 标准通用标记语言SGML
 - 移动图像专家组MPEG



MIME (4)

Type	Subtype	Description
Text	Plain	Unformatted text
	Enriched	Text including simple formatting commands
Image	Gif	Still picture in GIF format
	Jpeg	Still picture in JPEG format
Audio	Basic	Audible sound
Video	Mpeg	Movie in MPEG format
Application	Octet-stream	An uninterpreted byte sequence
	Postscript	A printable document in PostScript
Message	Rfc822	A MIME RFC 822 message
	Partial	Message has been split for transmission
	External-body	Message itself must be fetched over the net
Multipart	Mixed	Independent parts in the specified order
	Alternative	Same message in different formats
	Parallel	Parts must be viewed simultaneously
	Digest	Each part is a complete RFC 822 message

The MIME types and subtypes defined in RFC 2045.



MIME (5)

From: elinor@abcd.com
To: carolyn@xyz.com
MIME-Version: 1.0
Message-Id: <0704760941.AA00747@abcd.com>
Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
Subject: Earth orbits sun integral number of times

This is the preamble. The user agent ignores it. Have a nice day.

--qwertyuiopasdfghjklzxcvbnm
Content-Type: text/enriched

Happy birthday to you
Happy birthday to you
Happy birthday dear <bold> Carolyn </bold>
Happy birthday to you

--qwertyuiopasdfghjklzxcvbnm
Content-Type: message/external-body;
 access-type="anon-ftp";
 site="bicycle.abcd.com";
 directory="pub";
 name="birthday.snd"

content-type: audio/basic
content-transfer-encoding: base64
--qwertyuiopasdfghjklzxcvbnm--

A multipart message containing enriched and
audio alternatives.



Message Transfer

- SMTP — 简单邮件传输协议
 - 一个简单的ASCII协议
 - 端口号： 25
 - TCP连接
- 发送[fig 7-13]消息的简单对话-[see fig 7-14]



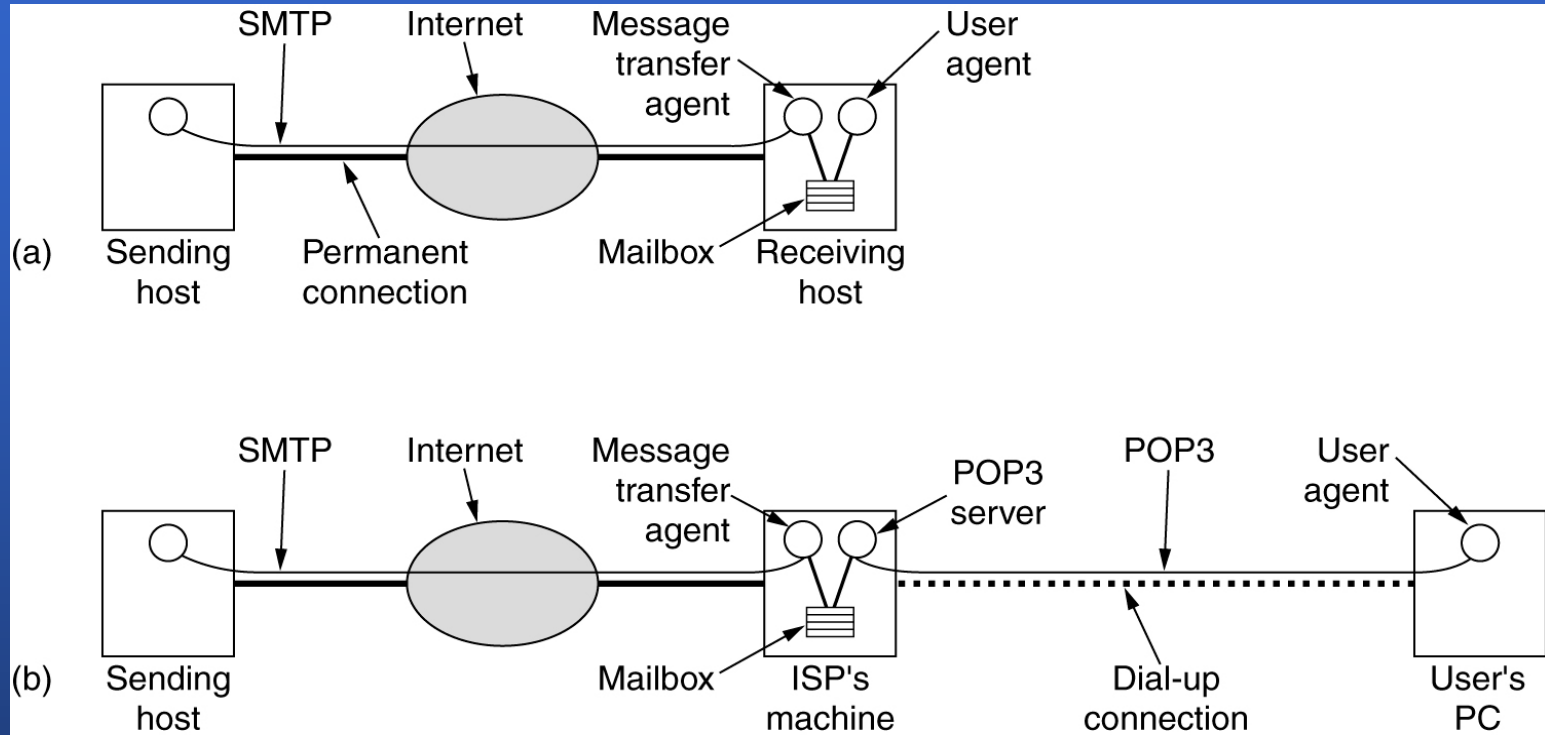
Message Transfer(2)

Transferring a
message from
elinore@abc.com
to
carolyn@xyz.com

```
S: 220 xyz.com SMTP service ready
C: HELO abcd.com
S: 250 xyz.com says hello to abcd.com
C: MAIL FROM: <elinor@abcd.com>
S: 250 sender ok
C: RCPT TO: <carolyn@xyz.com>
S: 250 recipient ok
C: DATA
S: 354 Send mail; end with "." on a line by itself
C: From: elinor@abcd.com
C: To: carolyn@xyz.com
C: MIME-Version: 1.0
C: Message-Id: <0704760941.AA00747@abcd.com>
C: Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
C: Subject: Earth orbits sun integral number of times
C:
C: This is the preamble. The user agent ignores it. Have a nice day.
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: text/enriched
C:
C: Happy birthday to you
C: Happy birthday to you
C: Happy birthday dear <bold> Carolyn </bold>
C: Happy birthday to you
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: message/external-body;
C:     access-type="anon-ftp";
C:     site="bicycle.abcd.com";
C:     directory="pub";
C:     name="birthday.snd"
C:
C: content-type: audio/basic
C: content-transfer-encoding: base64
C: --qwertyuiopasdfghjklzxcvbnm
C: .
S: 250 message accepted
C: QUIT
S: 221 xyz.com closing connection
```



7.2.5 Final Delivery



(a) Sending and reading mail when the receiver has a permanent Internet connection and the user agent runs on the same machine as the message transfer agent. (b) Reading e-mail when the receiver has a dial-up connection to an ISP.

Final Delivery(2)

- 最后的递交 – 解决非因特网上机器的消息递交
 - 邮局协议POP3
 - 从远程邮箱中读取电子邮件的简单协议
 - 交互式电子邮件访问协议IMAP
 - 邮件服务器维护一个中心数据库，能够被各种其它机器访问
 - 分布式电子邮件系统协议DMSP
 - 将邮件从服务器上下载(download)到机器上，断开连接后再阅读
- 其它传输特性
 - 过滤器
 - 转发
 - 假期幽灵(vacation daemon)程序
 - 邮件自动处理机(robot), 处理收到的每封邮件



POP3

```
S: +OK POP3 server ready
C: USER carolyn
S: +OK
C: PASS vegetables
S: +OK login successful
C: LIST
S: 1 2505
S: 2 14302
S: 3 8122
S: .
C: RETR 1
S: (sends message 1)
C: DELE 1
C: RETR 2
S: (sends message 2)
C: DELE 2
C: RETR 3
S: (sends message 3)
C: DELE 3
C: QUIT
S: +OK POP3 server disconnecting
```

Using POP3 to fetch three messages.



IMAP

Feature	POP3	IMAP
Where is protocol defined?	RFC 1939	RFC 2060
Which TCP port is used?	110	143
Where is e-mail stored?	User's PC	Server
Where is e-mail read?	Off-line	On-line
Connect time required?	Little	Much
Use of server resources?	Minimal	Extensive
Multiple mailboxes?	No	Yes
Who backs up mailboxes?	User	ISP
Good for mobile users?	No	Yes
User control over downloading?	Little	Great
Partial message downloads?	No	Yes
Are disk quotas a problem?	No	Could be in time
Simple to implement?	Yes	No
Widespread support?	Yes	Growing

A comparison of POP3 and IMAP.



7.3 The World Wide Web

- Architectural Overview
- Static Web Documents
- Dynamic Web Documents
- HTTP – The HyperText Transfer Protocol
- Performance Enhancements
- The Wireless Web



the world's largest
particle physics centre

- 起源于1989年欧洲粒子物理研究室 (CERN)
 - 1989.3 最初计划, Tim Berners-Lee (CERN)
 - 1993.2 Mosaic发布 (Marc Andreessen, Univ. of Illinois)
 - 1994 CERN和MIT共同建立WWW集团
<http://www.w3.org>
 - 1995 Netscape (by Andreessen)
 - 1996 Internet Explorer 1.0
- 编写语言: HTML、CGI和Java



7.3.1 Architectural Overview

- Web由庞大的、世界范围的文档组成—页面 (page)
- 超文本(hypertext)
- 浏览器(browser): Mosaic、Netscape、IE
- 超链接(hyperlink)
- 书签(bookmark)
- 超媒体(hypermedia)
- TCP连接 (port number: 80)



Architectural Overview(2)

WELCOME TO THE UNIVERSITY OF EAST PODUNK'S WWW HOME PAGE

- Campus Information
 - ☐ [Admissions information](#)
 - ☐ [Campus map](#)
 - ☐ [Directions to campus](#)
 - ☐ [The UEP student body](#)
- Academic Departments
 - ☐ [Department of Animal Psychology](#)
 - ☐ [Department of Alternative Studies](#)
 - ☐ [Department of Microbiotic Cooking](#)
 - ☐ [Department of Nontraditional Studies](#)
 - ☐ [Department of Traditional Studies](#)

Webmaster@eastpodunk.edu

(a)

THE DEPARTMENT OF ANIMAL PSYCHOLOGY

- [Information for prospective majors](#)
- Personnel
 - ☐ [Faculty members](#)
 - ☐ [Graduate students](#)
 - ☐ [Nonacademic staff](#)
- [Research Projects](#)
- [Positions available](#)
- Our most popular courses
 - ☐ [Dealing with herbivores](#)
 - ☐ [Horse management](#)
 - ☐ [Negotiating with your pet](#)
 - ☐ [User-friendly doghouse construction](#)
- [Full list of courses](#)

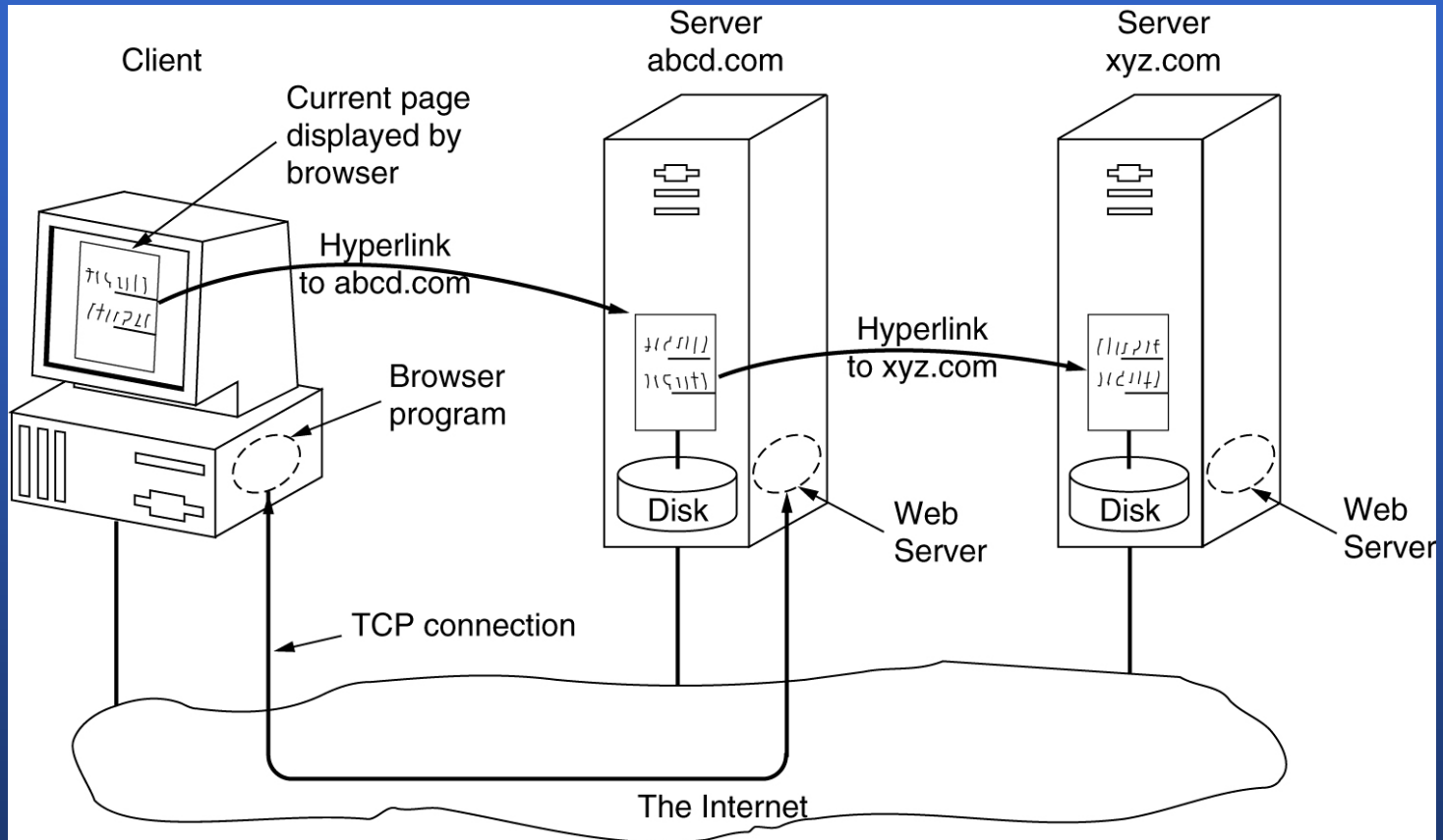
Webmaster@animalpsyc.eastpodunk.edu

(b)

(a) A Web page (b) The page reached by clicking on Department of Animal Psychology.

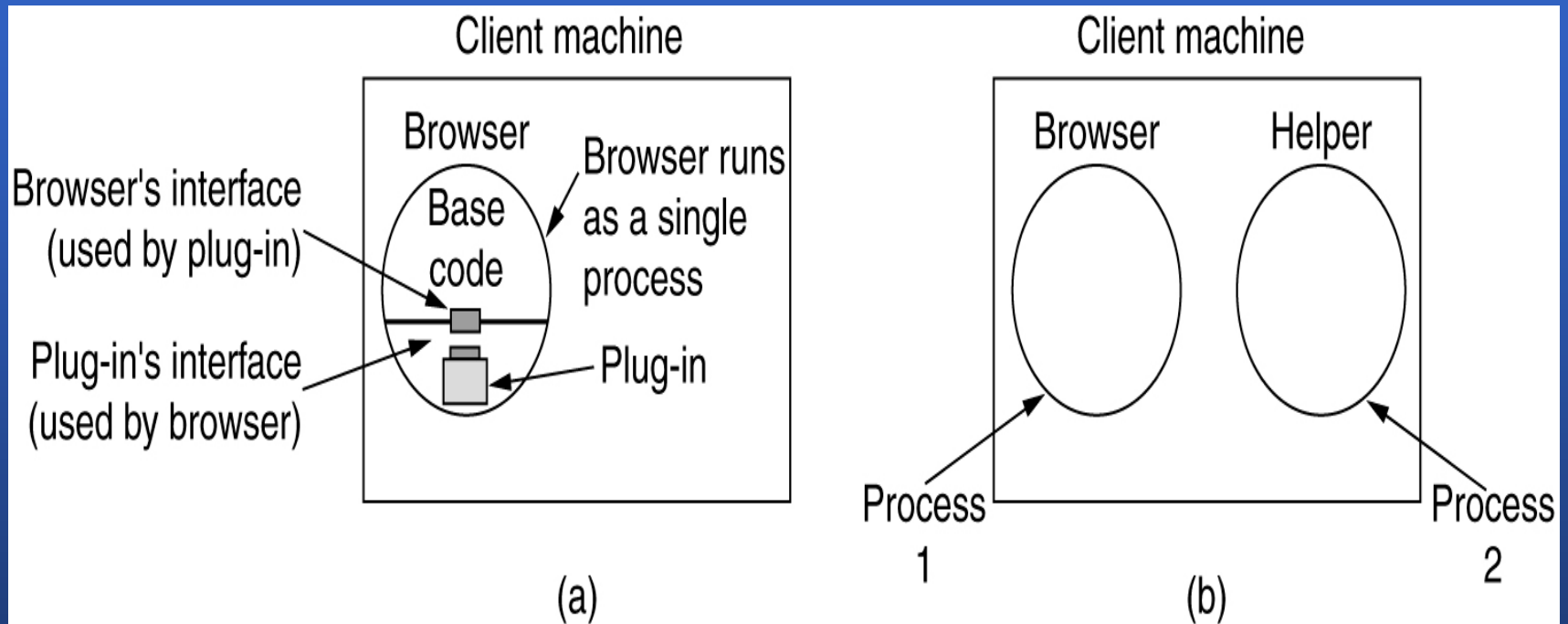


Architectural Overview (3)



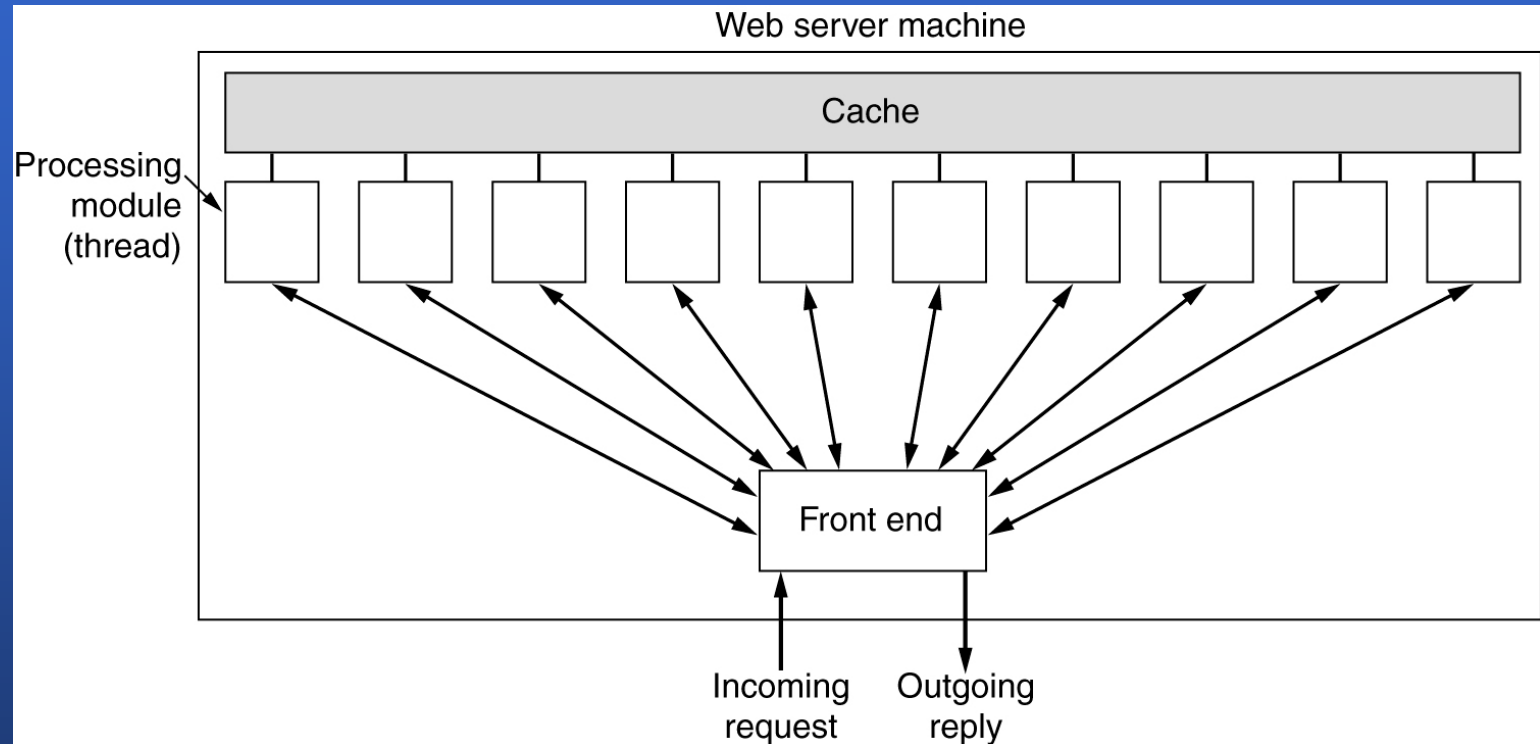
The parts of the Web model.

The Client Side



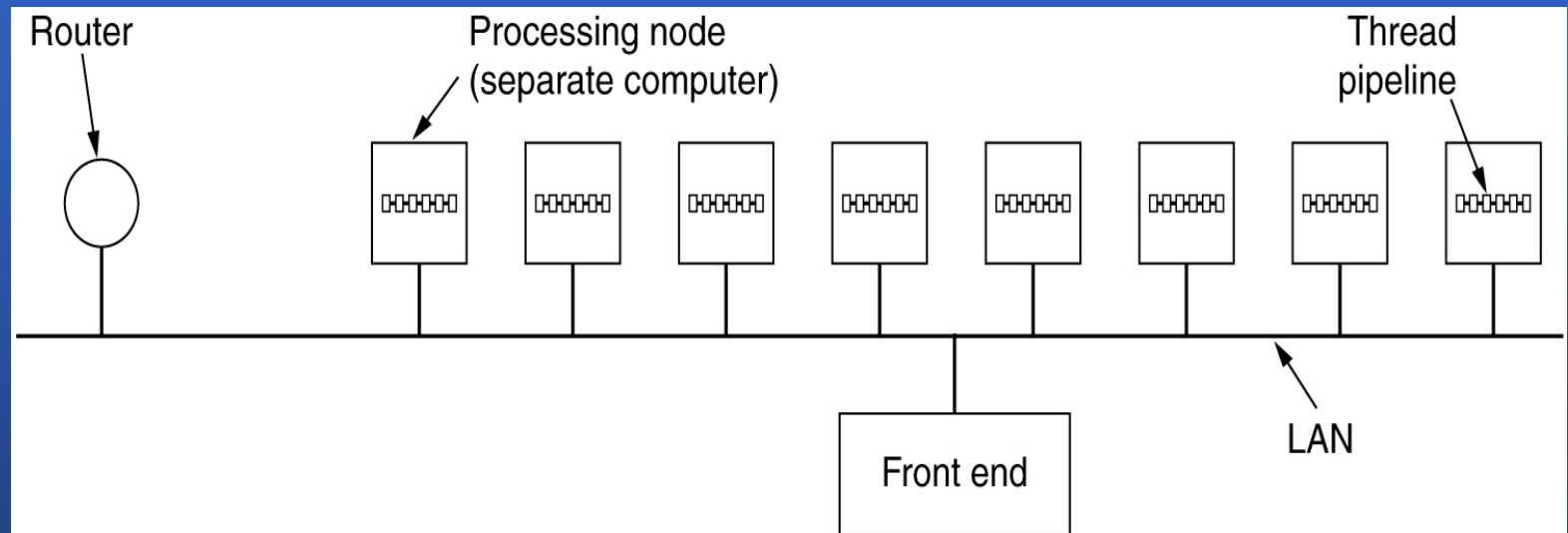
(a) A browser plug-in. **(b)** A helper application.
两种运行方式

The Server Side



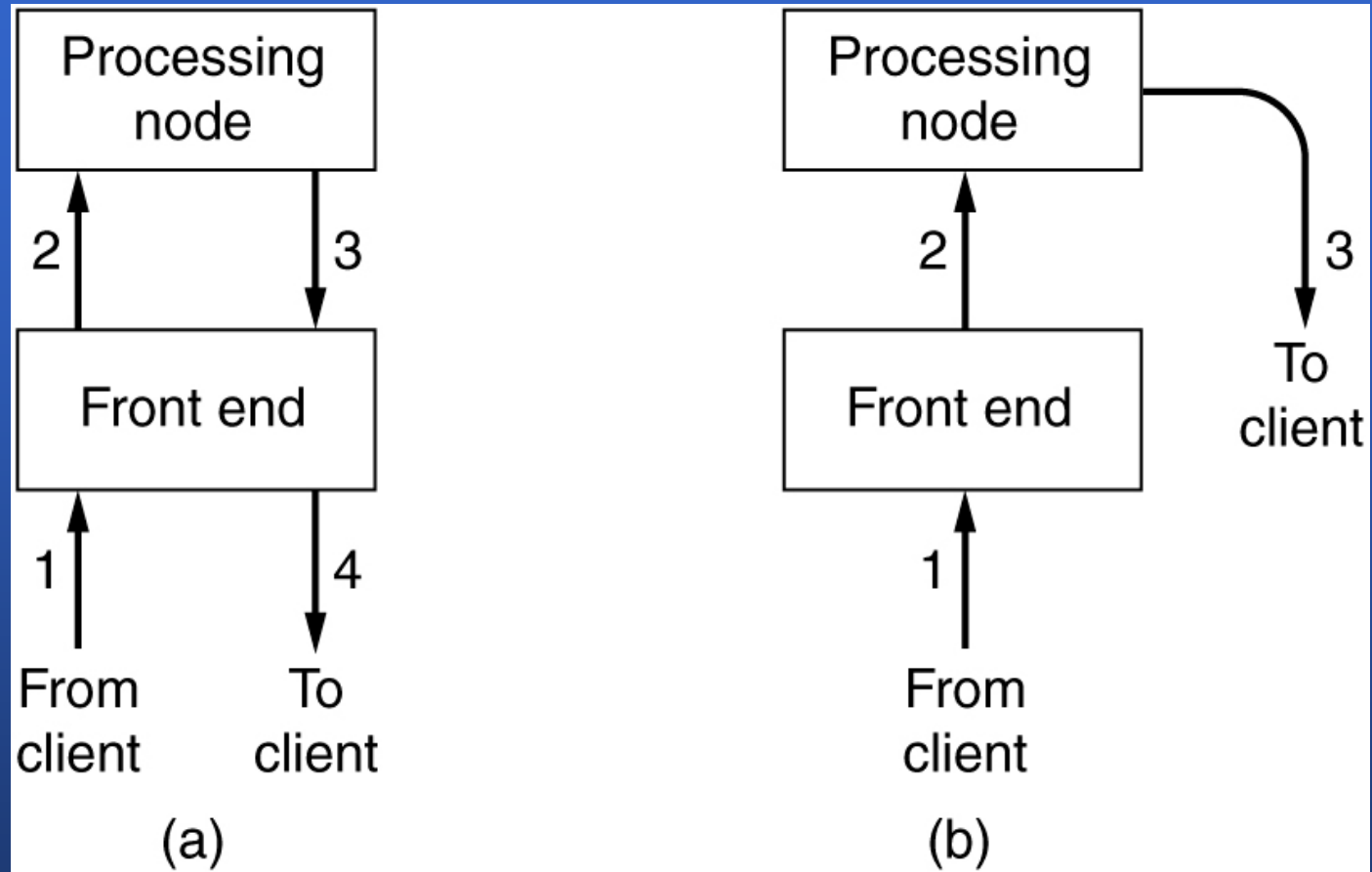
A multithreaded Web server with a front end and processing modules.

The Server Side (2)



A server farm.

The Server Side (3)



(a) Normal request-reply message sequence.

(b) Sequence when TCP handoff(传递) is used.

URLs – Uniform Resource Locators

- URL consists of 3 parts:
 - the **protocol** (http), the **DNS name** of the host (www.zju.edu.cn), and the **file name** (index.html)
- URL中常用的协议:

Name	Used for	Example
http	Hypertext (HTML)	http://www.cs.vu.nl/~ast/
ftp	FTP	ftp://ftp.cs.vu.nl/pub/minix/README
file	Local file	file:///usr/suzanne/prog.c
news	Newsgroup	news:comp.os.minix
news	News article	news:AA0134223112@cs.utah.edu
gopher	Gopher	gopher://gopher.tc.umn.edu/11/Libraries
mailto	Sending e-mail	mailto:JohnUser@acm.org
telnet	Remote login	telnet://www.w3.org:80



Statelessness and Cookies

- Web is stateless
 - No register
 - How does the server keep track of the contents of the client accessing?
 - To solve the problem, Netscape devised the technique: **cookies!**
- Cookies
 - When a client requests a Web page, the server can **supply additional information** along with the requested page
 - Cookies are just files or strings, not executable programs
 - Cookies could contain virus information in data file



Statelessness and Cookies

- A cookie may contain up to 5 fields:
 - Domain: where the cookie comes from
 - Path: which part of the server's file tree may use the cookie
 - Content: format as *name=value*
 - Expire: **non-persistent** cookie and **persistent** cookie
 - Secure: for e-commerce, banking or other secure applications
- Some examples of cookies:

Domain	Path	Content	Expires	Secure
toms-casino.com	/	CustomerID=497793521	15-10-02 17:00	Yes
joes-store.com	/	Cart=1-00501;1-07031;2-13721	11-10-02 14:22	No
aportal.com	/	Prefs=Stk:SUNW+ORCL;Spt:Jets	31-12-10 23:59	No
sneaky.com	/	UserID=3627239101	31-12-12 23:59	No



7.3.2 Static Web Documents

● HTML – HyperText Markup Language

```
<html>
<head><title> AMALGAMATED WIDGET, INC. </title> </head>
<body><h1> Welcome to AWI's Home Page</h1>
 <br>
We are so happy that you have chosen to visit <b> Amalgamated Widget's </b>
homepage.We hope<i> you </i> will find all the information you need here.
<p>Below we have links to information about our many fine products.
You can order electronically (by WWW), by telephone, or by fax. </p>
<hr>
<h2> Product information </h2>
<ul>
<li> <a href="http://widget.com/products/big"> Big widgets</a>
<li> <a href="http://widget.com/products/little"> Little widgets </a>
</ul>
<h2> Telephone numbers</h2>
<ul>
<li> By telephone:1-800-WIDGETS
<li> By fax: 1-415-765-4321
</ul>
</body>
</html>
```

(a)

Welcome to AWI's Home Page



We are so happy that you have chosen to visit **Amalgamated Widget's** home page. We hope you will find all the information you need here.

Below we have links to information about our many fine products. You can order electronically (by WWW), by telephone, or by FAX.

Product Information

- [Big widgets](#)
- [Little widgets](#)

Telephone numbers

- 1-800-WIDGETS
- 1-415-765-4321

(a) The HTML for a sample Web page. (b) The formatted page.



HTML (2)

Tag	Description
<html> ... </html>	Declares the Web page to be written in HTML
<head> ... </head>	Delimits the page's head
<title> ... </title>	Defines the title (not displayed on the page)
<body> ... </body>	Delimits the page's body
<h <i>n</i> > ... </h <i>n</i> >	Delimits a level <i>n</i> heading
 ... 	Set ... in boldface
<i> ... </i>	Set ... in italics
<center> ... </center>	Center ... on the page horizontally
...	Brackets an unordered (bulleted) list
...	Brackets a numbered list
	Starts a list item (there is no)
 	Forces a line break here
<p>	Starts a paragraph
<hr>	Inserts a Horizontal rule
	Displays an image here
 ... 	Defines a hyperlink

A selection of common HTML tags. some can have additional parameters.

Forms

```
<html>
<head> <title> A sample page with a table </title> </head>
<body>
<table border=1 rules=all>
<caption> Some Differences between HTML Versions </caption>
<col align=left>
<col align=center>
<col align=center>
<col align=center>
<col align=center>
<tr> <th>Item <th>HTML 1.0 <th>HTML 2.0 <th>HTML 3.0 <th>HTML 4.0 </tr>
<tr> <th> Hyperlinks <td> x <td> x <td> x <td> x </tr>
<tr> <th> Images <td> x <td> x <td> x <td> x </tr>
<tr> <th> Lists <td> x <td> x <td> x <td> x </tr>
<tr> <th> Active Maps and Images <td> &nbsp;&nbsp;&nbsp;& <td> x <td> x <td> x </tr>
<tr> <th> Forms <td> &nbsp;&nbsp;& <td> x <td> x <td> x </tr>
<tr> <th> Equations <td> &nbsp;&nbsp;& <td> &nbsp;&nbsp;& <td> x <td> x </tr>
<tr> <th> Toolbars <td> &nbsp;&nbsp;& <td> &nbsp;&nbsp;& <td> x <td> x </tr>
<tr> <th> Tables <td> &nbsp;&nbsp;& <td> &nbsp;&nbsp;& <td> x <td> x </tr>
<tr> <th> Accessibility features <td> &nbsp;&nbsp;& <td> &nbsp;&nbsp;& <td> &nbsp;&nbsp;& <td> x </tr>
<tr> <th> Object embedding <td> &nbsp;&nbsp;& <td> &nbsp;&nbsp;& <td> &nbsp;&nbsp;& <td> x </tr>
<tr> <th> Scripting <td> &nbsp;&nbsp;& <td> &nbsp;&nbsp;& <td> &nbsp;&nbsp;& <td> x </tr>
</table>
</body>
</html>
```

(a)

(a) An HTML table.

(b) A possible rendition of this table.

Some Differences between HTML Versions

Item	HTML 1.0	HTML 2.0	HTML 3.0	HTML 4.0
Hyperlinks	x	x	x	x
Images	x	x	x	x
Lists	x	x	x	x
Active Maps and Images		x	x	x
Forms		x	x	x
Equations			x	x
Toolbars			x	x
Tables			x	x
Accessibility features				x
Object embedding				x
Scripting				x

(b)



Forms (2)

(a) The HTML for an order form.

(b) The formatted page.

```
<html>
<head> <title> AWI CUSTOMER ORDERING FORM </title> </head>
<body>
<h1> Widget Order Form </h1>
<form ACTION="http://widget.com/cgi-bin/widgetorder" method=POST>
<p> Name <input name="customer" size=46> </p>
<p> Street Address <input name="address" size=40> </p>
<p> City <input name="city" size=20> State <input name="state" size=4>
Country <input name="country" size=10> </p>
<p> Credit card # <input name="cardno" size=10>
Expires <input name="expires" size=4>
M/C <input name="cc" type=radio value="mastercard">
VISA <input name="cc" type=radio value="visacard"> </p>
<p> Widget size Big <input name="product" type=radio value="expensive">
Little <input name="product" type=radio value="cheap">
Ship by express courier <input name="express" type=checkbox> </p>
<p><input type=submit value="submit order"> </p>
Thank you for ordering an AWI widget, the best widget money can buy!
</form>
</body>
</html>
```

(a)

Widget Order Form

Name

Street address

City State Country

Credit card # Expires M/C ☐ Visa ☐

Widget size Big ☐ Little ☐ Ship by express courier ☐

Thank you for ordering an AWI widget, the best widget money can buy!



Forms (3)

```
customer=John+Doe&address=100+Main+St.&city=White+Plains&  
state=NY&country=USA&cardno=1234567890&expires=6/98&cc=mastercard&  
product=cheap&express=on
```

A possible response from the browser to the server with information filled in by the user.

XML and XSL

To enhance the ability for HTML to be structured for automated processing, W3C develop two new languages: XML and XSL

XML– eXtensible Markup Language (to describe Web content in a structured way)

XSL-- eXtensible Style Language (to describe the formatting of the content)

XML and XSL are still static

```
<?xml version="1.0" ?>
<?xml-stylesheet type="text/xsl" href="b5.xsl"?>
<book_list>
  <book>
    <title> Computer Networks, 4/e </title>
    <author> Andrew S. Tanenbaum </author>
    <year> 2003 </year>
  </book>
  <book>
    <title> Modern Operating Systems, 2/e </title>
    <author> Andrew S. Tanenbaum </author>
    <year> 2001 </year>
  </book>
  <book>
    <title> Structured Computer Organization, 4/e </title>
    <author> Andrew S. Tanenbaum </author>
    <year> 1999 </year>
  </book>
</book_list>
```

A simple Web page in XML.



XML and XSL (2)

```
<?xml version='1.0'?>
<xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform" version="1.0">
<xsl:template match="/">

<html>
<body>

<table border="2">
  <tr>
    <th> Title</th>
    <th> Author</th>
    <th> Year </th>
  </tr>

  <xsl:for-each select="book_list/book">
    <tr>
      <td> <xsl:value-of select="title"/> </td>
      <td> <xsl:value-of select="author"/> </td>
      <td> <xsl:value-of select="year"/> </td>
    </tr>
  </xsl:for-each>
</table>

</body>
</html>
</xsl:template>
</xsl:stylesheet>
```

A style sheet
in XSL.

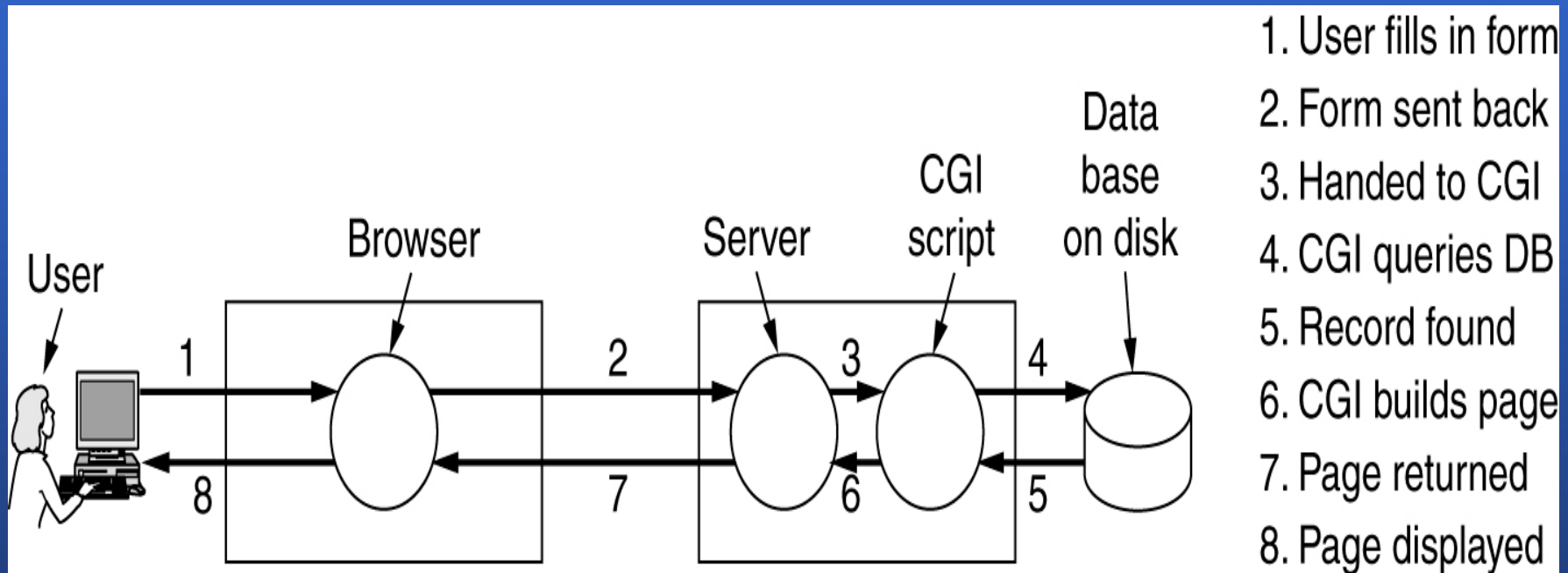


XHTML –eXtended HyperText Markup Language

- In order to fitting to the use of the Web-enabled devices, such wireless and handheld PDA-type devices
- Limited memory and small size display
- XHTML – a new Web standard
- Some differences between XHTML and HTML4



7.3.3 Dynamic Web Documents



Steps in processing the information from an HTML form.

Server-Side Dynamic Web Page Generation

- Four techniques (languages) to generate dynamic content on the server side (**dynamic HTML**):
 - (1) **CGI(Common Gateway Interface)**
 - A traditional way
 - 在服务器 (cgi-bin) 上运行的一段脚本程序，最终将运行结果传回客户端
 - Perl
 - (2) **PHP(PHP: Hypertext Preprocessor)**
 - To embed little scripts inside HTML pages and have them be executed by the server itself to generate the page
 - A popular language for writing these scripts
 - `<?php ... ?>` tag contained in normal HTML file
 - File extension *php*



Server-Side Dynamic Web Page Generation (2)

- (3) JSP (Java Server Pages)
 - Similar to PHP
 - Written in the Java language instead of in PHP
 - File extension *.jsp*
- (4) ASP (Active Server Pages)
 - Microsoft's version of PHP and JSP
 - File extension *.asp*



Dynamic Web Documents (3)

```
<html>
<body>

<h2> This is what I know about you </h2>
<?php echo $HTTP_USER_AGENT ?>

</body>
</html>
```

A sample HTML page with embedded PHP.

(file: www.abcd.com/test.php)



Dynamic Web Documents (4)

```
<html>
<body>
<form action="action.php" method="post">
<p> Please enter your name: <input type="text" name="name"> </p>
<p> Please enter your age: <input type="text" name="age"> </p>
<input type="submit">
</form>
</body>
</html>
```

(a)

```
<html>
<body>
<h1> Reply: </h1>
Hello <?php echo $name; ?>.
Prediction: next year you will be <?php echo $age + 1; ?>
</body>
</html>
```

(b)

```
<html>
<body>
<h1> Reply: </h1>
Hello Barbara.
Prediction: next year you will be 25
</body>
</html>
```

(c)

(a) A Web page containing a form. (b) A PHP script for handling the output of the form. (c) Output from the PHP script when the inputs are "Barbara" and 24 respectively.



Client-Side Dynamic Web Page Generation

Use of JavaScript
for processing a
form.

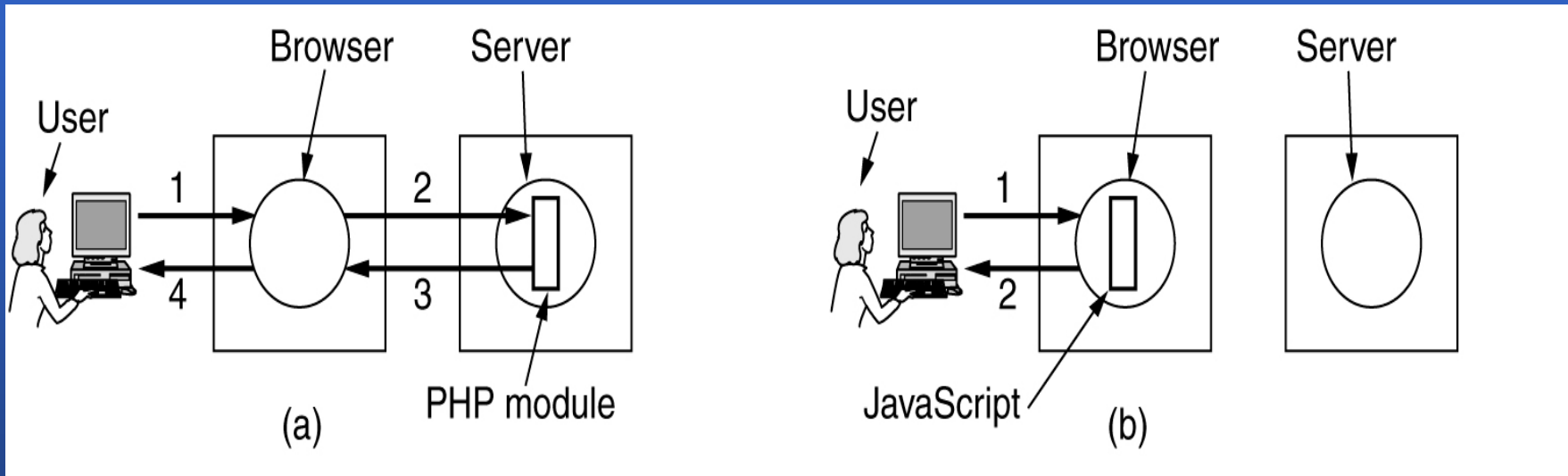
Using tag *<script>*

```
<head>
<script language="javascript" type="text/javascript">
function response(test form) {
    var person = test form.name.value;
    var years = eval(test form.age.value) + 1;
    document.open();
    document.writeln("<html> <body>");
    document.writeln("Hello " + person + ".<br>");
    document.writeln("Prediction: next year you will be " + years + ".");
    document.writeln("</body> </html>");
    document.close();
}
</script>
</head>

<body>
<form>
Please enter your name: <input type="text" name="name">
<p>
Please enter your age: <input type="text" name="age">
<p>
<input type="button" value="submit" onclick="response(this.form)">
</form>
</body>
</html>
```



Client-Side Dynamic Web Page Generation (2)



(a) Server-side scripting with PHP.

(b) Client-side scripting with JavaScript.

Client-Side Dynamic Web Page Generation (3)

```
<html>
<head>
<script language="javascript" type="text/javascript">
function response(test_form) {
    function factorial(n) {if (n == 0) return 1; else return n * factorial(n - 1);}
    var r = eval(test_form.number.value);    // r = typed in argument
    document.myform.mytext.value = "Here are the results.\n";
    for (var i = 1; i <= r; i++)              // print one line from 1 to r
        document.myform.mytext.value += (i + "! = " + factorial(i) + "\n");
    }
</script>
</head>

<body>
<form name="myform">
Please enter a number: <input type="text" name="number">
<input type="button" value="compute table of factorials" onclick="response(this.form)">
<p>
<textarea name="mytext" rows=25 cols=50> </textarea>
</form>
</body>
</html>
```

A JavaScript program for computing and printing factorials.



Client-Side Dynamic Web Page Generation (4)

```
<html>
<head>
<script language="javascript" type="text/javascript">
if (!document.myurl) document.myurl = new Array();
document.myurl[0] = "http://www.cs.vu.nl/~ast/im/kitten.jpg";
document.myurl[1] = "http://www.cs.vu.nl/~ast/im/puppy.jpg";
document.myurl[2] = "http://www.cs.vu.nl/~ast/im/bunny.jpg";
function pop(m) {
    var urx = "http://www.cs.vu.nl/~ast/im/cat.jpg";
    popupwin = window.open(document.myurl[m],"mywind","width=250,height=250");
}
</script>
</head>

<body>
<p> <a href="#" onmouseover="pop(0); return false;" > Kitten </a> </p>
<p> <a href="#" onmouseover="pop(1); return false;" > Puppy </a> </p>
<p> <a href="#" onmouseover="pop(2); return false;" > Bunny </a> </p>
</body>
</html>
```

An interactive Web page that responds to mouse movement.

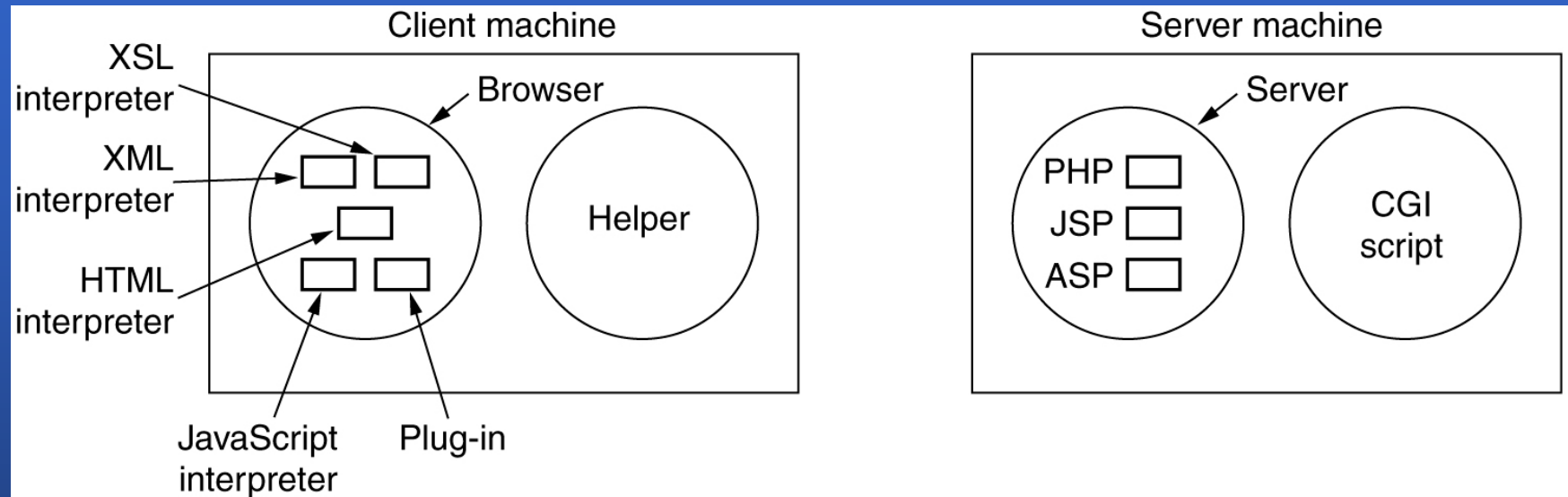
Client-Side Dynamic Web Page Generation (5)

The various ways to generate and display content

- JavaScript
 - Easier to write
- Java applet
 - `<applet> ... </applet>`
 - Execute faster
- ActiveX controls
 - Microsoft's answer to Sun's Java applets
 - Run fastest of all



Client-Side Dynamic Web Page Generation (6)



The various ways to generate and display content.

7.3.4 HTTP– Hyper Text Transfer Protocol

- ASCII request
- RFC 822 MIME-like response
- TCP Connection: **port 80** on server
- HTTP 1.1, supports **persistent connections**
 - To establish a TCP connection, send a request and get a response, and then send additional requests and get additional responses



HTTP Methods

Method	Description
GET	Request to read a Web page
HEAD	Request to read a Web page's header
PUT	Request to store a Web page
POST	Append to a named resource (e.g., a Web page)
DELETE	Remove the Web page
TRACE	Echo the incoming request
CONNECT	Reserved for future use
OPTIONS	Query certain options

The **built-in HTTP request methods**.



HTTP Methods (2)

Code	Meaning	Examples
1xx	Information	100 = server agrees to handle client's request
2xx	Success	200 = request succeeded; 204 = no content present
3xx	Redirection	301 = page moved; 304 = cached page still valid
4xx	Client error	403 = forbidden page; 404 = page not found
5xx	Server error	500 = internal server error; 503 = try again later

The **status code** response groups.



HTTP Message Headers

Header	Type	Contents
User-Agent	Request	Information about the browser and its platform
Accept	Request	The type of pages the client can handle
Accept-Charset	Request	The character sets that are acceptable to the client
Accept-Encoding	Request	The page encodings the client can handle
Accept-Language	Request	The natural languages the client can handle
Host	Request	The server's DNS name
Authorization	Request	A list of the client's credentials
Cookie	Request	Sends a previously set cookie back to the server
Date	Both	Date and time the message was sent
Upgrade	Both	The protocol the sender wants to switch to
Server	Response	Information about the server
Content-Encoding	Response	How the content is encoded (e.g., gzip)
Content-Language	Response	The natural language used in the page
Content-Length	Response	The page's length in bytes
Content-Type	Response	The page's MIME type
Last-Modified	Response	Time and date the page was last changed
Location	Response	A command to the client to send its request elsewhere
Accept-Ranges	Response	The server will accept byte range requests
Set-Cookie	Response	The server wants the client to save a cookie

Some HTTP message headers.



Example HTTP Usage

The start of the output
of
www.ietf.org/rfc.ht
ml

```
Trying 4.17.168.6...
Connected to www.ietf.org.
Escape character is '^]'.
HTTP/1.1 200 OK
Date: Wed, 08 May 2002 22:54:22 GMT
Server: Apache/1.3.20 (Unix) mod_ssl/2.8.4 OpenSSL/0.9.5a
Last-Modified: Mon, 11 Sep 2000 13:56:29 GMT
ETag: "2a79d-c8b-39bce48d"
Accept-Ranges: bytes
Content-Length: 3211
Content-Type: text/html
X-Pad: avoid browser bug

<html>
<head>
<title>IETF RFC Page</title>

<script language="javascript">
function url() {
  var x = document.form1.number.value
  if (x.length == 1) {x = "000" + x }
  if (x.length == 2) {x = "00" + x }
  if (x.length == 3) {x = "0" + x }
  document.form1.action = "/rfc/rfc" + x + ".txt"
  document.form1.submit
}
</script>

</head>
```

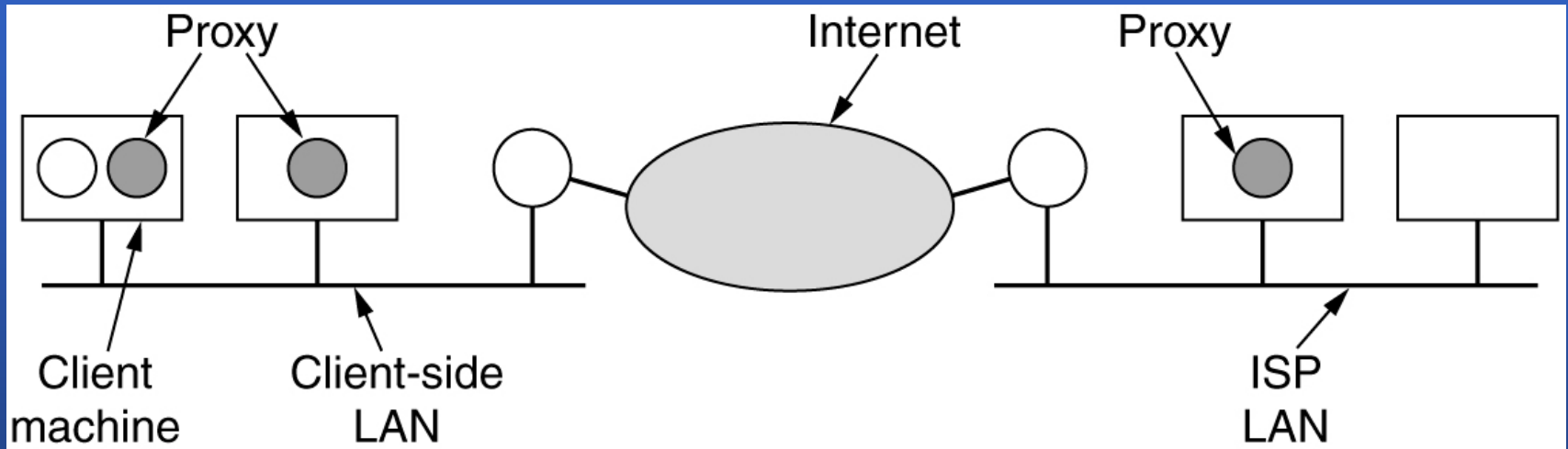


7.3.5 Performance Enhancements

- Caching
- Server Replication (or called Mirroring)
- Server replication
- Content delivery networks (CDN)
 - Content providers
 - ICP vs. ISP



Caching— Client-side technique



Hierarchical caching with three proxies.

Mirroring— Server-side technique

- Server Replication, also called Mirroring
- Static
- Flash crowds (瞬间拥挤)



Content Delivery Networks

```
<html>
<head> <title> Furry Video </title> </head>
<body>
<h1> Furry Video's Product List </h1>
<p> Click below for free samples. </p>

<a href="bears.mpg"> Bears Today </a> <br>
<a href="bunnies.mpg"> Funny Bunnies </a> <br>
<a href="mice.mpg"> Nice Mice </a> <br>
</body>
</html>
```

(a)

```
<html>
<head> <title> Furry Video </title> </head>
<body>
<h1> Furry Video's Product List </h1>
<p> Click below for free samples. </p>

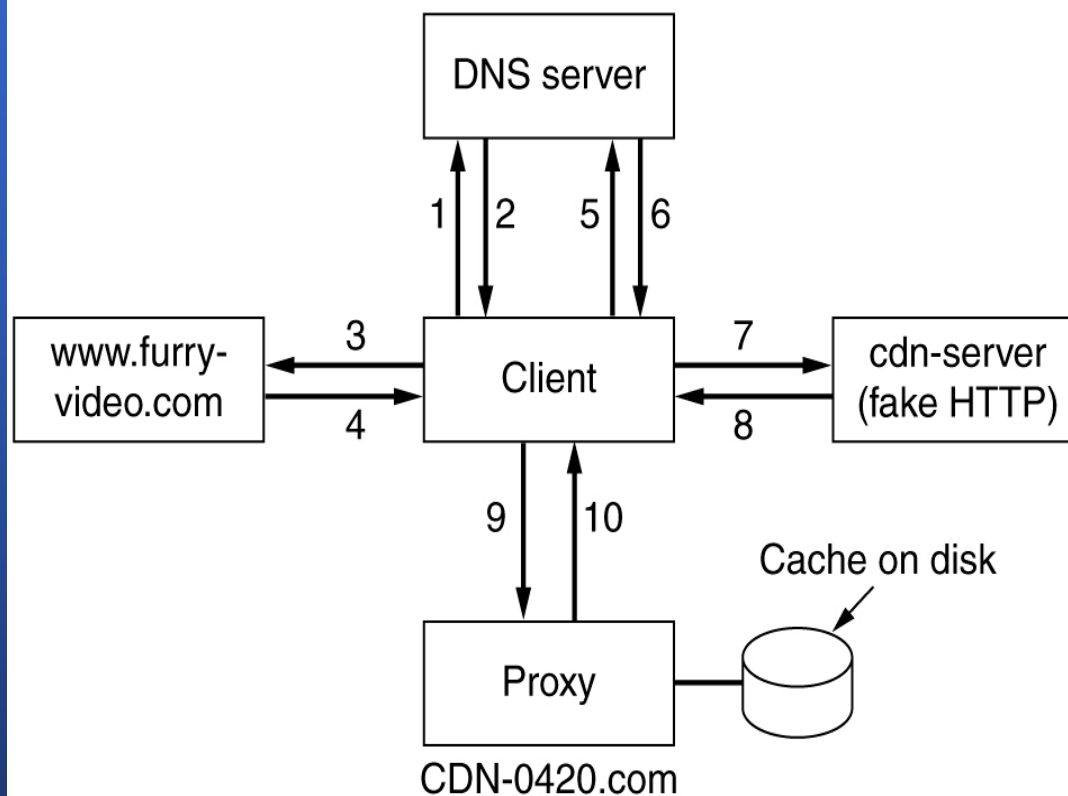
<a href="http://cdn-server.com/furryvideo/bears.mpg"> Bears Today </a> <br>
<a href="http://cdn-server.com/furryvideo/bunnies.mpg"> Funny Bunnies </a> <br>
<a href="http://cdn-server.com/furryvideo/mice.mpg"> Nice Mice </a> <br>
</body>
</html>
```

(b)

(a) Original Web page. **(b)** Same page after transformation.



Content Delivery Networks(2)



1. Look up www.furryvideo.com
2. Furry's IP address returned
3. Request HTML page from Furry
4. HTML page returned
5. After click, look up cdn-server.com
6. IP address of cdn-server returned
7. Ask cdn-server for bears.mpg
8. Client told to redirect to CDN-0420.com
9. Request bears.mpg
10. Cached file bears.mpg returned

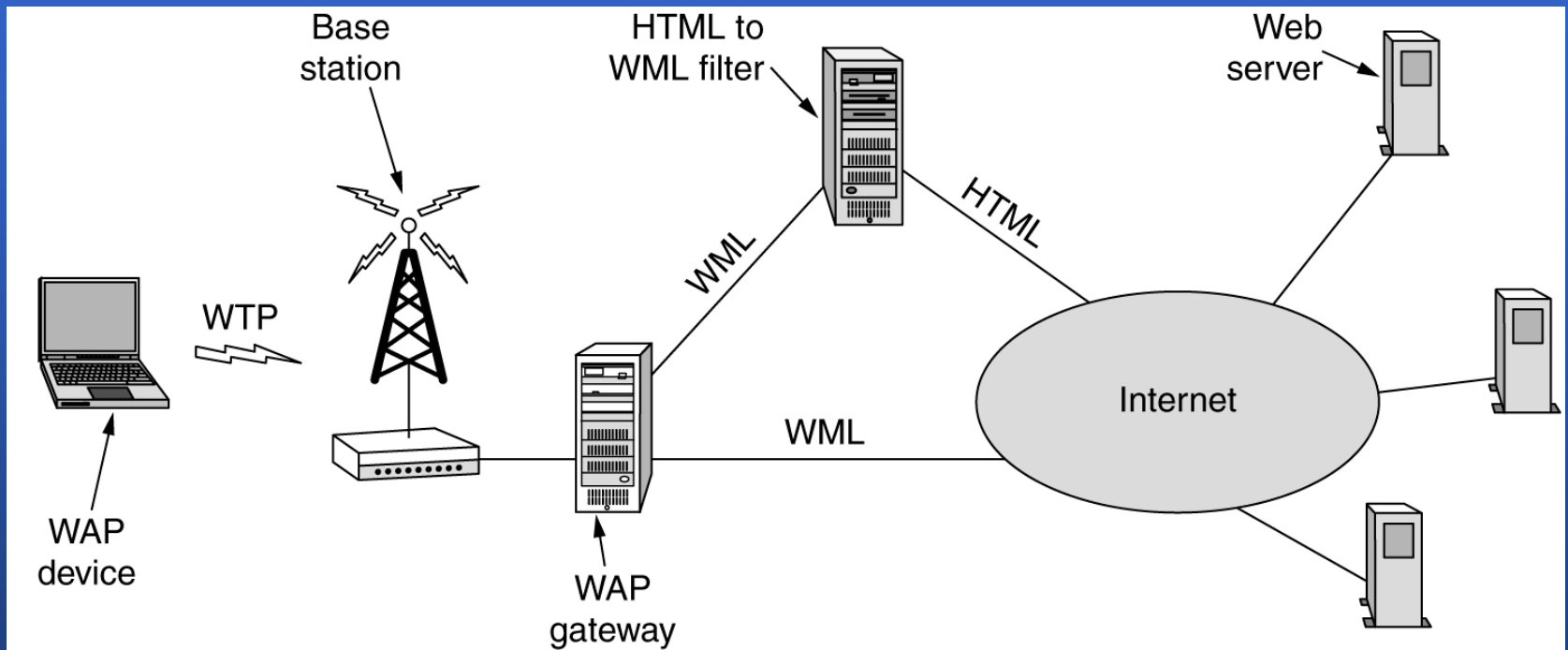
Steps in looking up a URL when a CDN is used.

7.3.6 WAP – The Wireless Application Protocol

Wireless application environment (WAE)
Wireless session protocol (WSP)
Wireless transaction protocol (WTP)
Wireless transport layer security (WTLS)
Wireless datagram protocol (WDP)
Bearer layer (GSM, CDMA, D-AMPS, GPRS, etc.)

The WAP protocol stack.

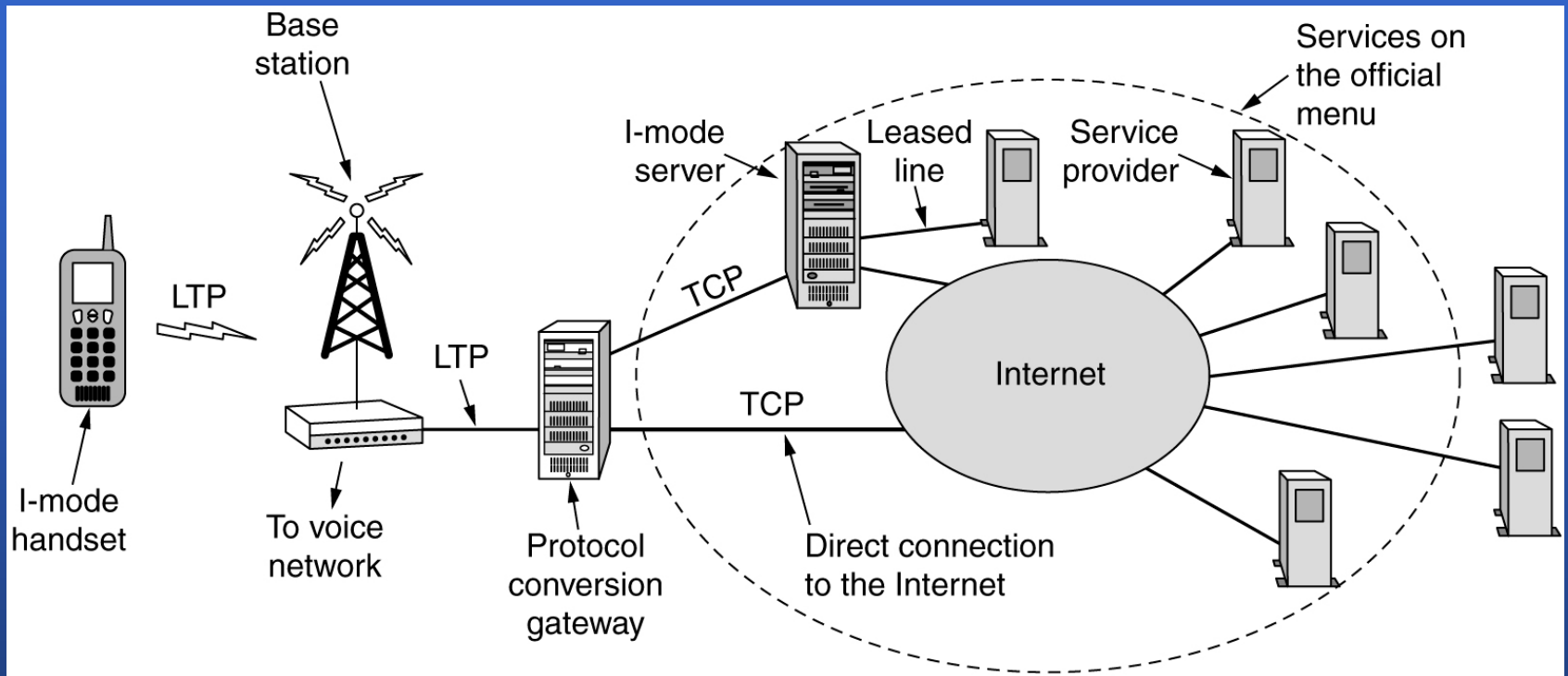
WAP (2)



The WAP architecture.

WML— Wireless Markup Language

I-Mode



Structure of the i-mode data network showing the transport protocols.

I-Mode – Information mode (Japan)

It was a huge success in Japan, but WAP 1.0 was failed

I-Mode (2)

User interaction module		
Plug-ins	cHTML interpreter	Java
Simple window manager		
Network communication		
Real-time operating system		

Structure of the i-mode software.

cHTML – compact HTML

I-Mode (3)

The time has come
the walrus said
to talk of many things.
Of shoes and ships and
sealing wax of c

(a)

The time has
come the walrus
said to talk of
many things. Of
shoes and ships
and sealing wax

(b)

Lewis Carroll meets a 16 x 16 screen.

I-Mode (4)

```
<html>
<body>
<h1> Select an option </h1>
<a href="messages.chnml" accesskey="1"> Check voicemail </a> <br>
<a href="mail.chnml" accesskey="2"> Check e-mail </a> <br>
<a href="games.chnml" accesskey="3"> Play a game </a>
</body>
</html>
```

An example of cHTML file.



Second-Generation Wireless Web

Feature	WAP	I-mode
What it is	Protocol stack	Service
Device	Handset, PDA, notebook	Handset
Access	Dial up	Always on
Underlying network	Circuit-switched	Two: circuit + packet
Data rate	9600 bps	9600 bps
Screen	Monochrome	Color
Markup language	WML (XML application)	cHTML
Scripting language	WMLscript	None
Usage charges	Per minute	Per packet
Pay for shopping	Credit card	Phone bill
Pictograms	No	Yes
Standardization	WAP forum open standard	NTT DoCoMo proprietary
Where used	Europe, Japan	Japan
Typical user	Businessman	Young woman

A comparison of first-generation WAP and i-mode.

Second-Generation Wireless Web (2)

New features of WAP 2.0.

- Push model as well as pull model.
- Support for integrating telephony into apps.
- Multimedia messaging.
- Inclusion of 264 pictograms.
- Interface to a storage device. (flash ROM)
- Support for plug-ins in the browser.

WAP 1.0 was failed



Second-Generation Wireless Web (3)

XHTML	
WSP	HTTP
WTP	TLS
WTLS	TCP
WDP	IP
Bearer layer	Bearer layer
WAP 1.0 protocol stack	WAP 2.0 protocol stack

WAP 2.0 supports two protocol stacks.

Second-Generation Wireless Web (4)

Module	Req.?	Function	Example tags
Structure	yes	Doc. structure	body, head, html, title
Text	yes	Information	br, code, dfn, em, hn, kbd, p, strong
Hypertext	yes	Hyperlinks	a
List	yes	Itemized lists	dl, dt, dd, ol, ul, li
Forms	No	Fill-in forms	form, input, label, option, textarea
Tables	No	Rectangular tables	caption, table, td, th, tr
Image	No	Pictures	img
Object	No	Applets, maps, etc.	object, param
Meta-information	No	Extra info	meta
Link	No	Similar to <a>	link
Base	No	URL starting point	base

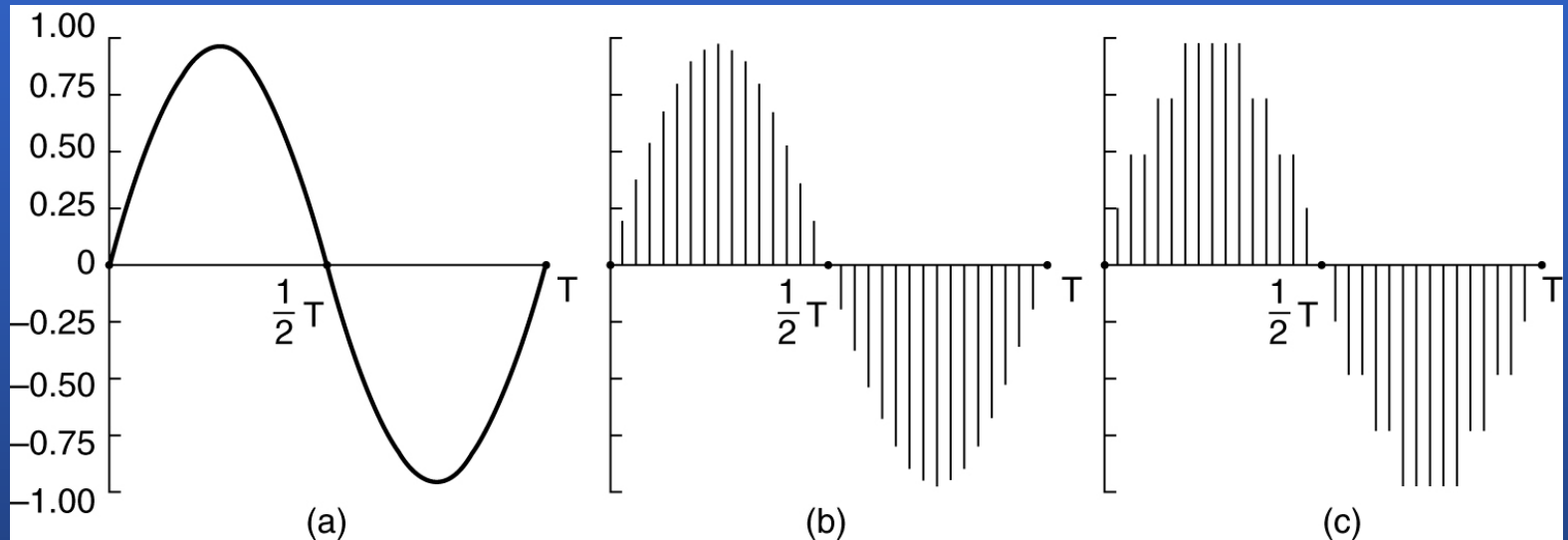
The XHTML Basic modules and tags.

7.4 Multimedia

- Introduction to Audio
- Audio Compression
- Streaming Audio
- Internet Radio
- Voice over IP
- Introduction to Video
- Video Compression
- Video on Demand
- The MBone – The Multicast Backbone

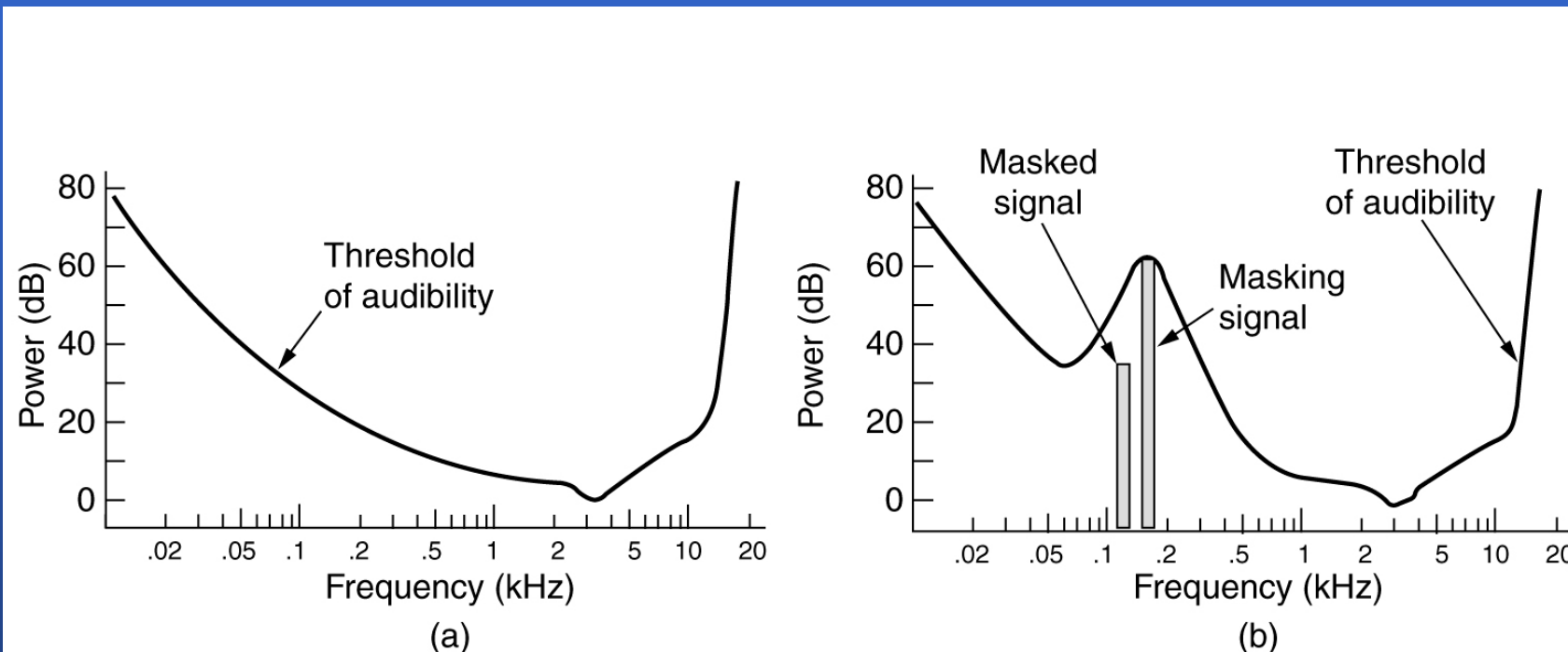


Introduction to Audio



(a) A sine wave. (b) Sampling the sine wave.
(c) Quantizing the samples to 4 bits.

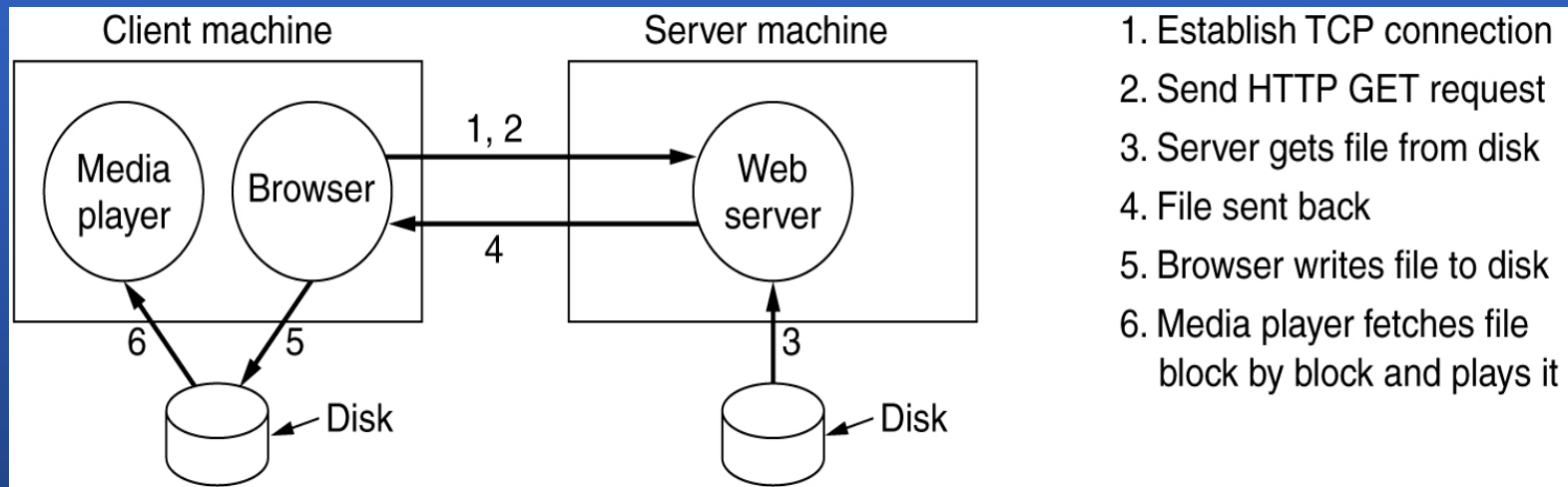
Audio Compression



(a) The threshold of audibility as a function of frequency.

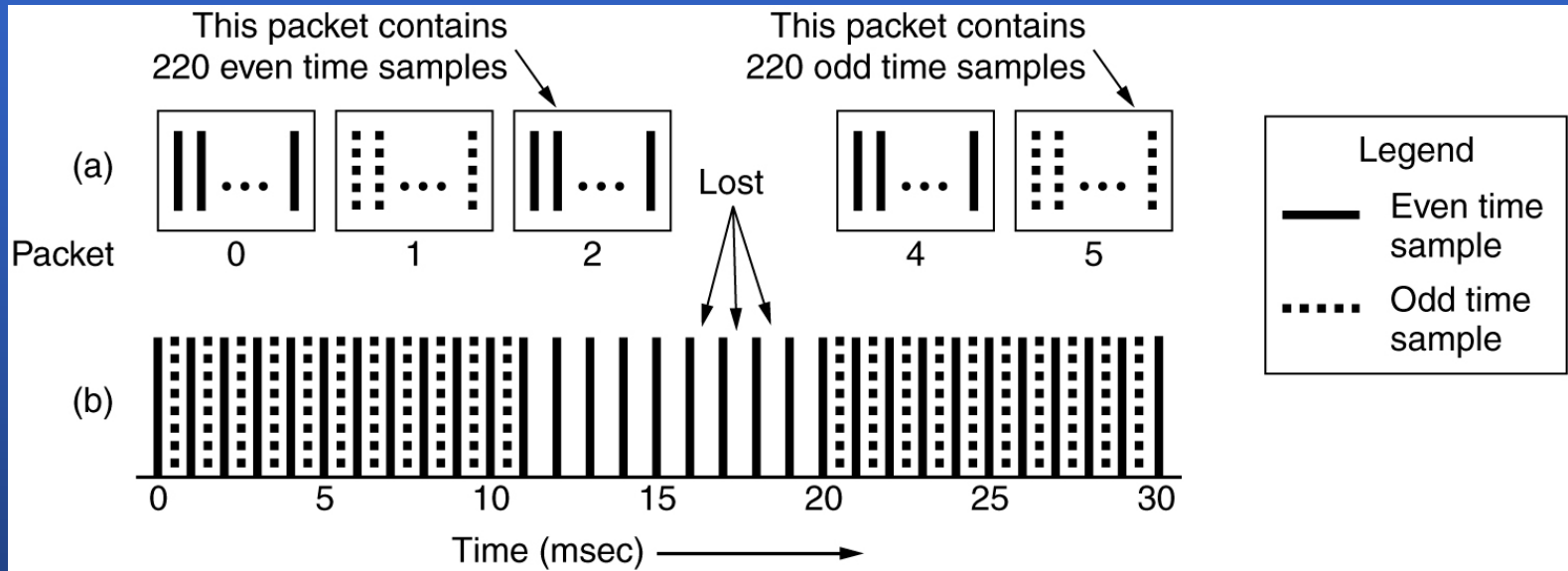
(b) The masking effect.

Streaming Audio



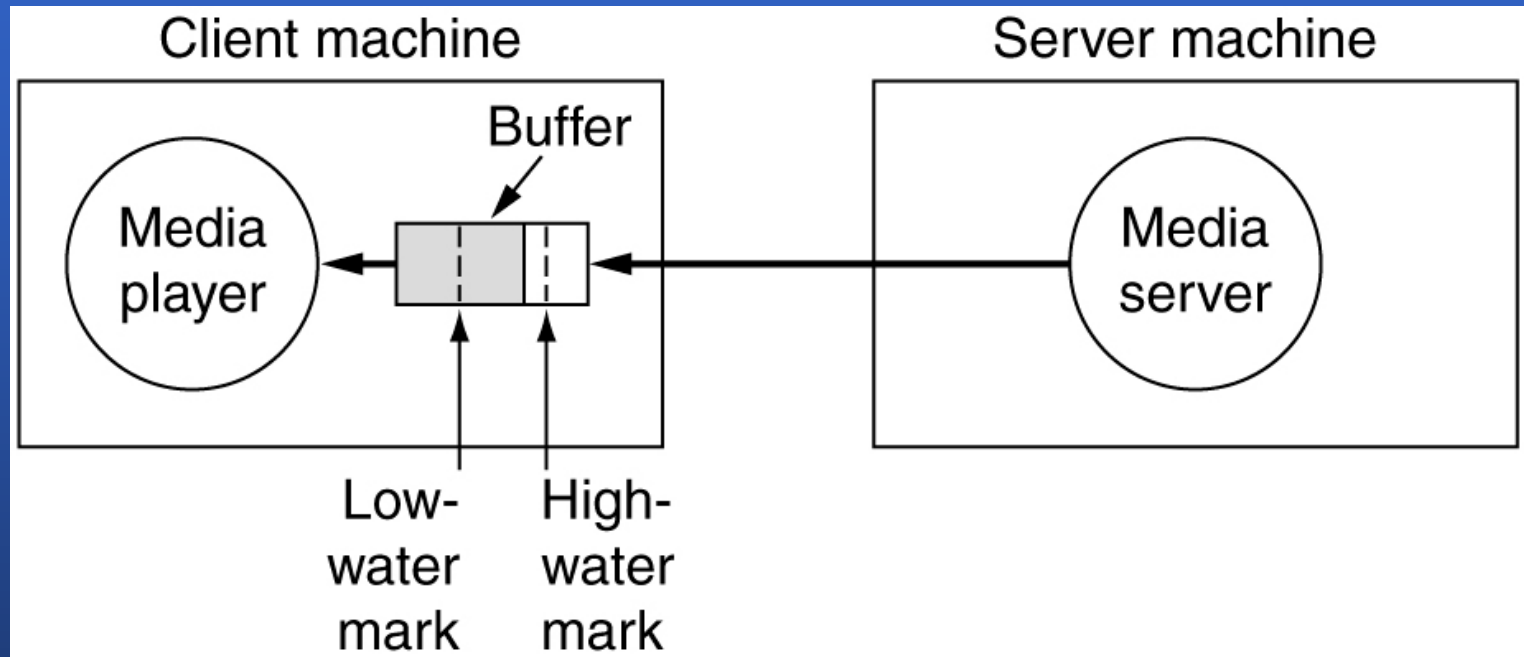
A straightforward way to implement clickable music on a Web page.

Streaming Audio (2)



When packets carry **alternate samples**, the loss of a packet reduces the temporal resolution rather than creating a gap in time.

Streaming Audio (3)



The media player **buffers** input from the media server and plays from the buffer rather than directly from the network.

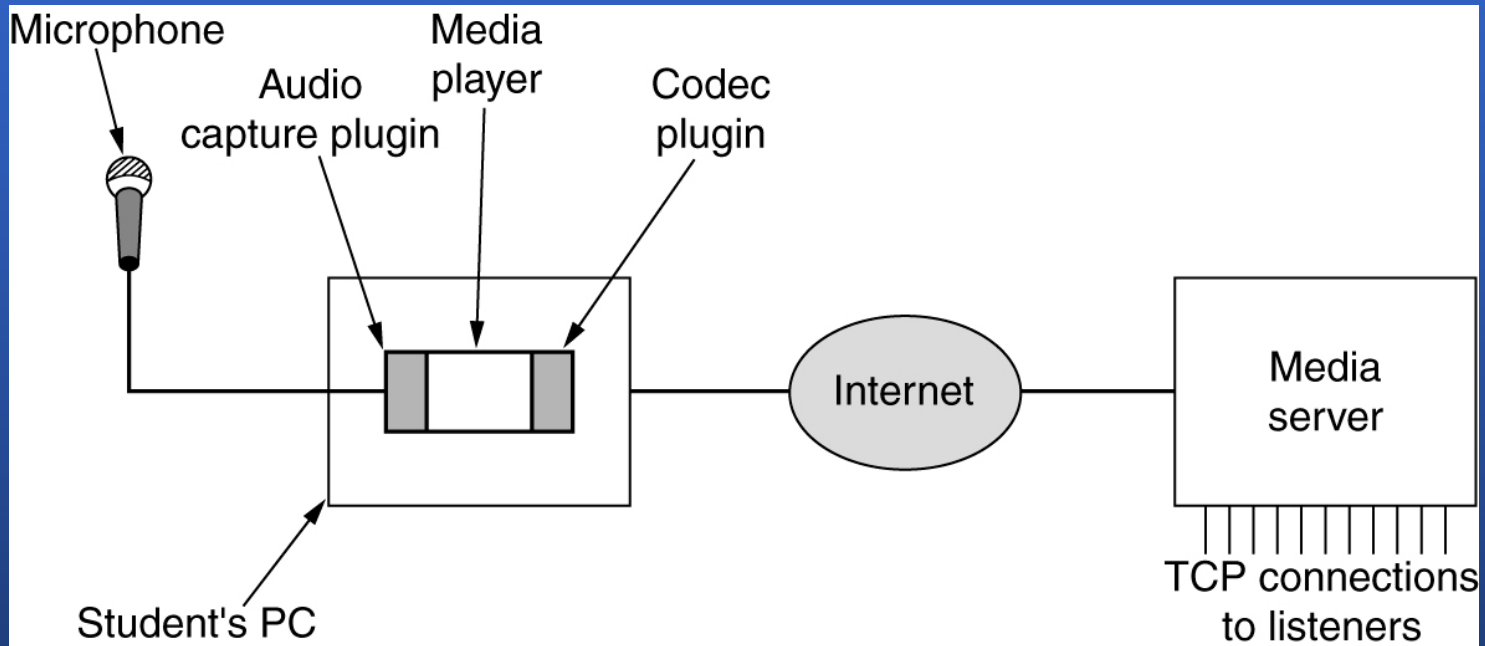
Streaming Audio (4)

Command	Server action
DESCRIBE	List media parameters
SETUP	Establish a logical channel between the player and the server
PLAY	Start sending data to the client
RECORD	Start accepting data from the client
PAUSE	Temporarily stop sending data
TEARDOWN	Release the logical channel

RTSP commands from the player to the server.

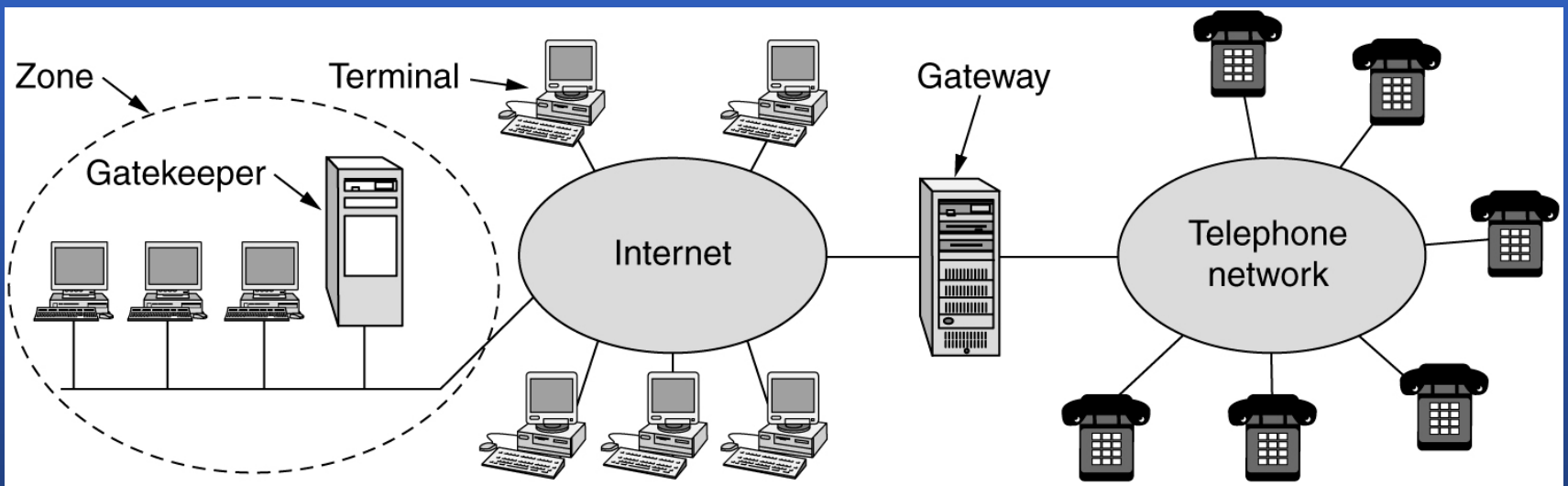


Internet Radio



A student radio station.

Voice over IP



The H.323 architectural model for Internet telephony. [designed by ITU]

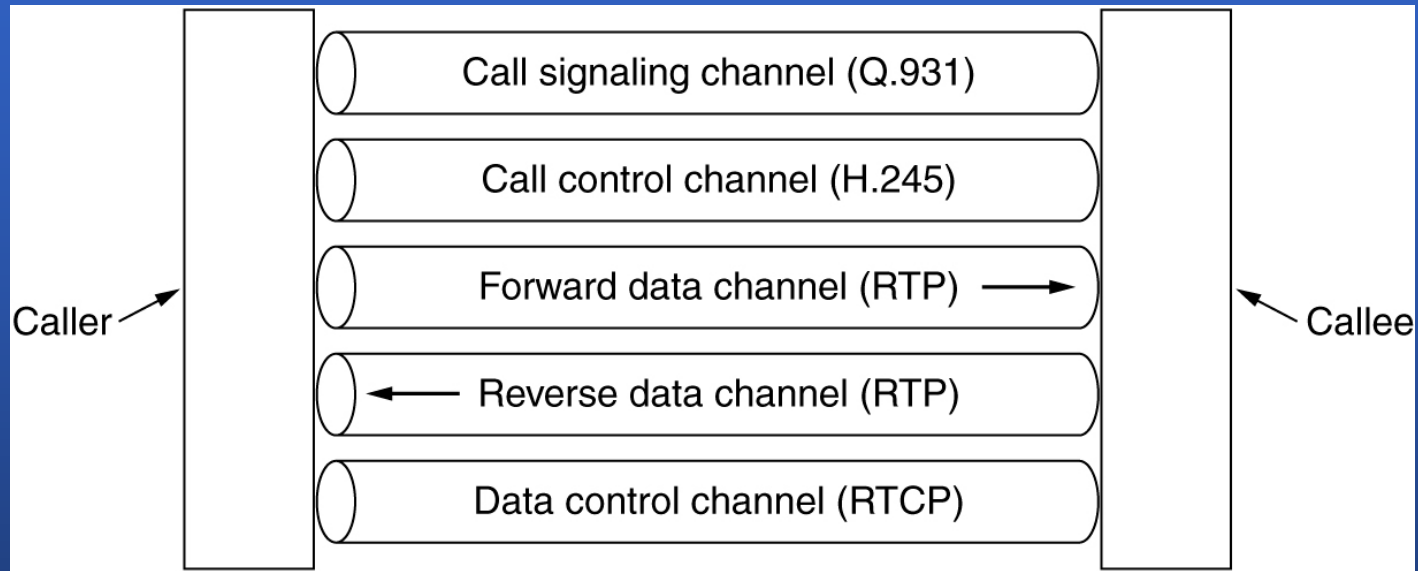
Voice over IP (2)

Speech	Control			
G.7xx	RTCP	H.225 (RAS)	Q.931 (Call signaling)	H.245 (Call control)
RTP				
UDP			TCP	
IP				
Data link protocol				
Physical layer protocol				

The H.323 protocol stack.



Voice over IP (3)



Logical channels between the caller and callee during a call.

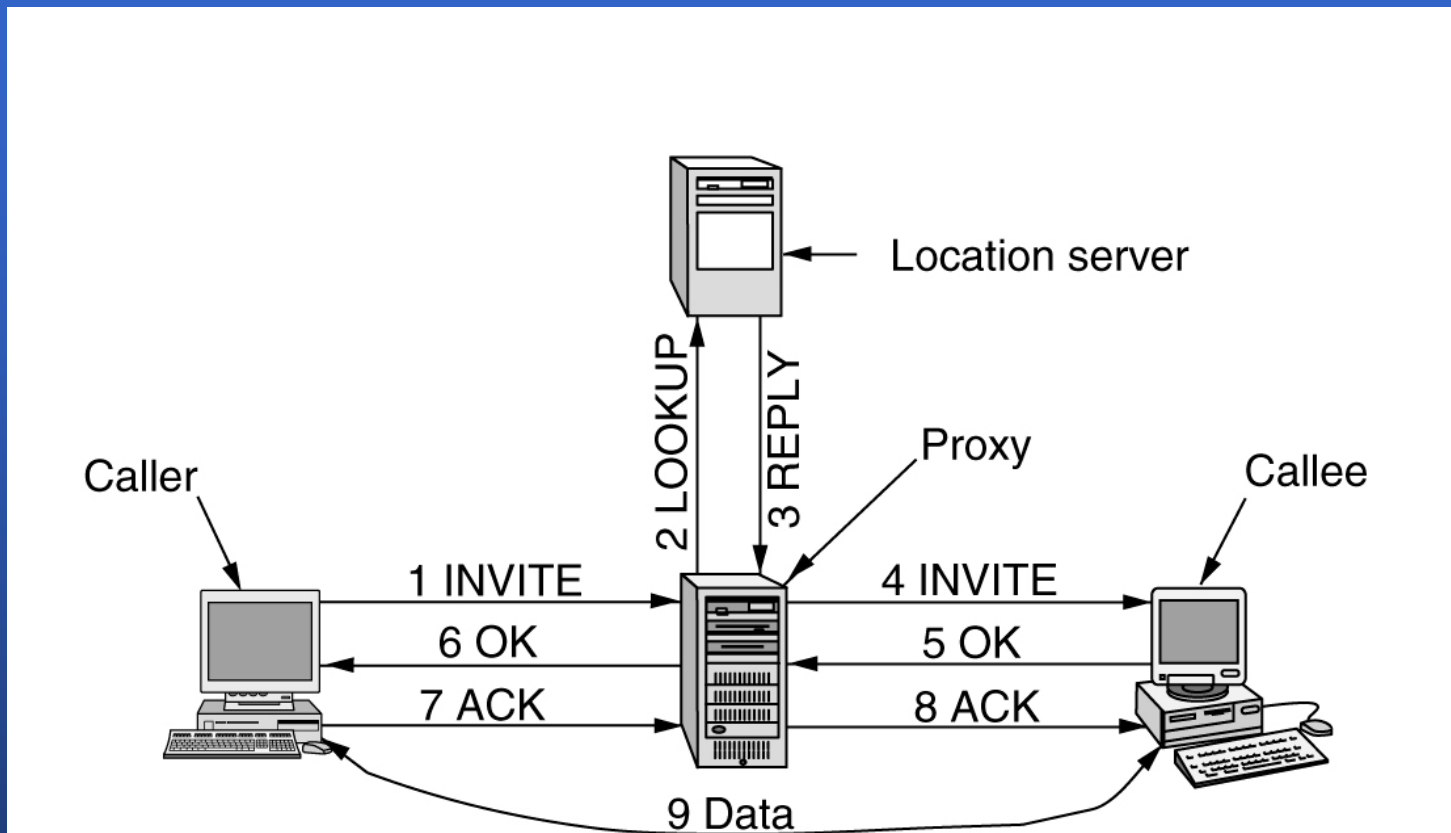
SIP – The Session Initiation Protocol

Method	Description
INVITE	Request initiation of a session
ACK	Confirm that a session has been initiated
BYE	Request termination of a session
OPTIONS	Query a host about its capabilities
CANCEL	Cancel a pending request
REGISTER	Inform a redirection server about the user's current location

The SIP methods defined in the core specification. [designed by IETF]



SIP (2)



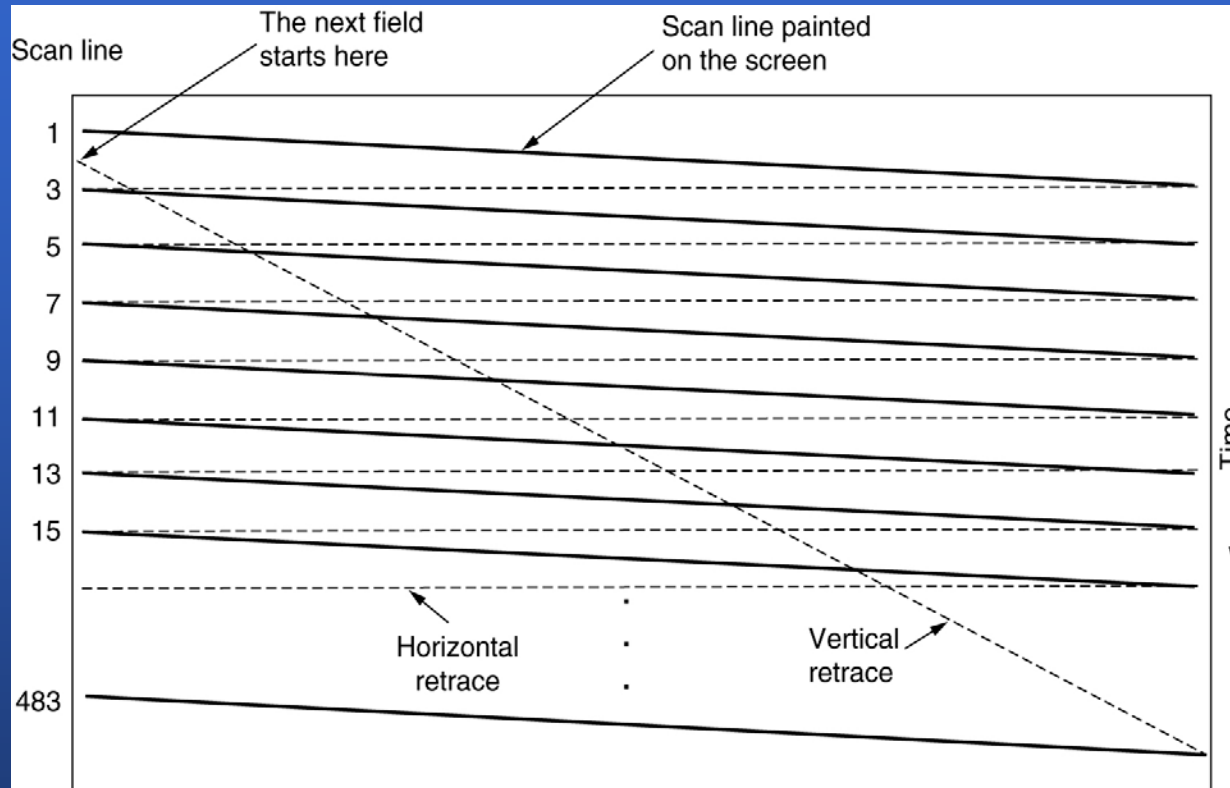
Use a proxy and redirection servers with SIP.

Comparison of H.323 and SIP

Item	H.323	SIP
Designed by	ITU	IETF
Compatibility with PSTN	Yes	Largely
Compatibility with Internet	No	Yes
Architecture	Monolithic	Modular
Completeness	Full protocol stack	SIP just handles setup
Parameter negotiation	Yes	Yes
Call signaling	Q.931 over TCP	SIP over TCP or UDP
Message format	Binary	ASCII
Media transport	RTP/RTCP	RTP/RTCP
Multiparty calls	Yes	Yes
Multimedia conferences	Yes	No
Addressing	Host or telephone number	URL
Call termination	Explicit or TCP release	Explicit or timeout
Instant messaging	No	Yes
Encryption	Yes	Yes
Size of standards	1400 pages	250 pages
Implementation	Large and complex	Moderate
Status	Widely deployed	Up and coming

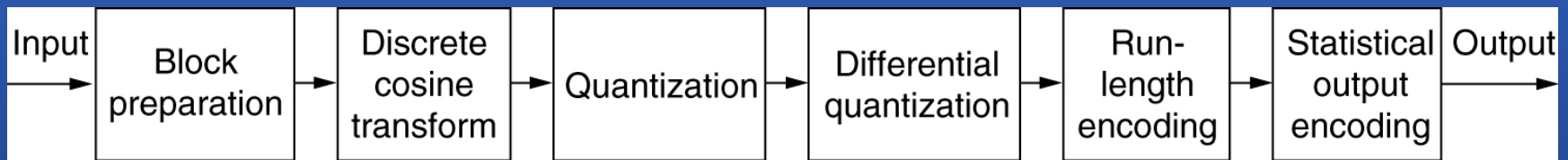


Video Analog Systems



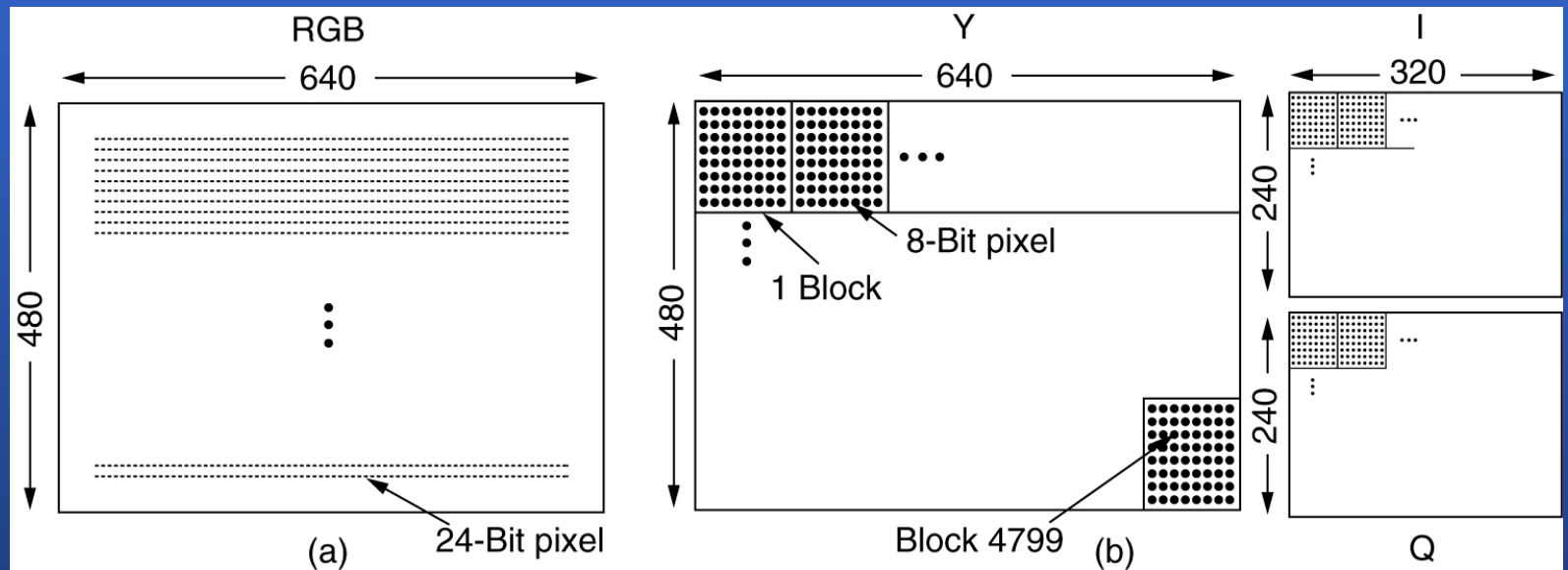
The scanning pattern used for NTSC video and television.

The JPEG Standard



The operation of JPEG in lossy sequential mode.

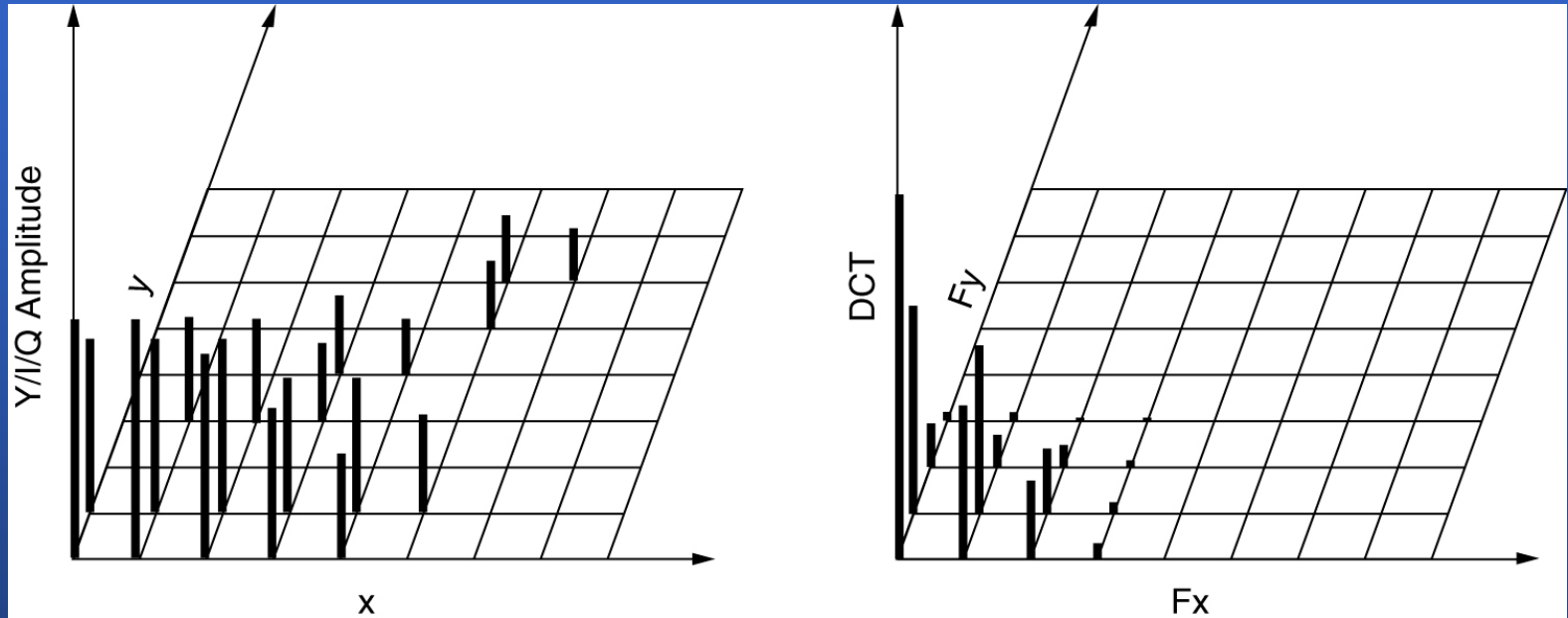
The JPEG Standard (2)



(a) RGB input data.

(b) After block preparation.

The JPEG Standard (3)



(a)

(b)

(a) One block of the Y matrix.

(b) The DTC coefficients.

The JPEG Standard (4)

DCT Coefficients								Quantization table								Quantized coefficients							
150	80	40	14	4	2	1	0	1	1	2	4	8	16	32	64	150	80	20	4	1	0	0	0
92	75	36	10	6	1	0	0	1	1	2	4	8	16	32	64	92	75	18	3	1	0	0	0
52	38	26	8	7	4	0	0	2	2	2	4	8	16	32	64	26	19	13	2	1	0	0	0
12	8	6	4	2	1	0	0	4	4	4	4	8	16	32	64	3	2	2	1	0	0	0	0
4	3	2	0	0	0	0	0	8	8	8	8	8	16	32	64	1	0	0	0	0	0	0	0
2	2	1	1	0	0	0	0	16	16	16	16	16	16	32	64	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	32	32	32	32	32	32	32	64	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	64	64	64	64	64	64	64	64	0	0	0	0	0	0	0	0

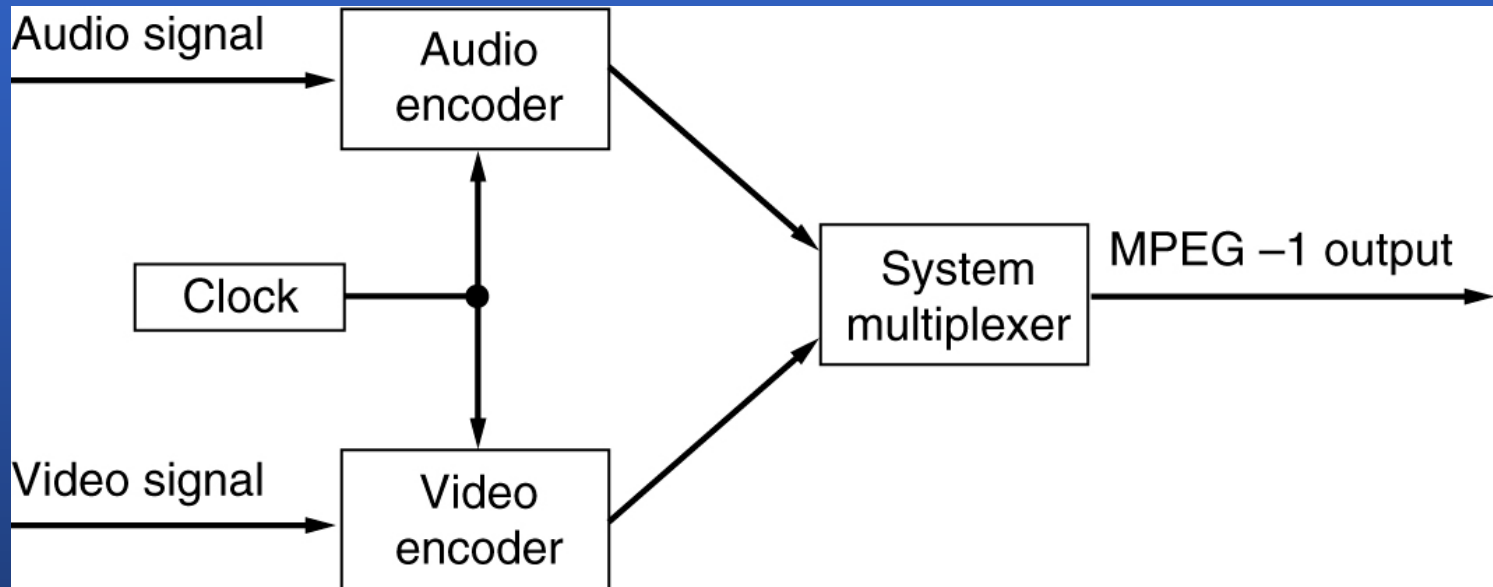
Computation of the quantized DTC coefficients.

The JPEG Standard (5)

150	80	20	4	1	0	0	0
92	75	18	3	1	0	0	0
26	19	13	2	1	0	0	0
3	2	2	1	0	0	0	0
1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

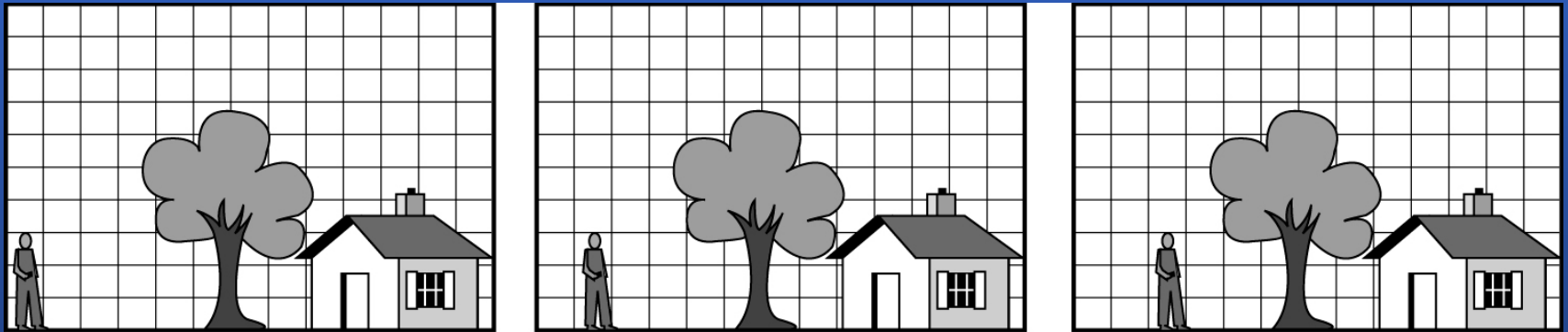
The order in which the quantized values are transmitted.

The MPEG Standard



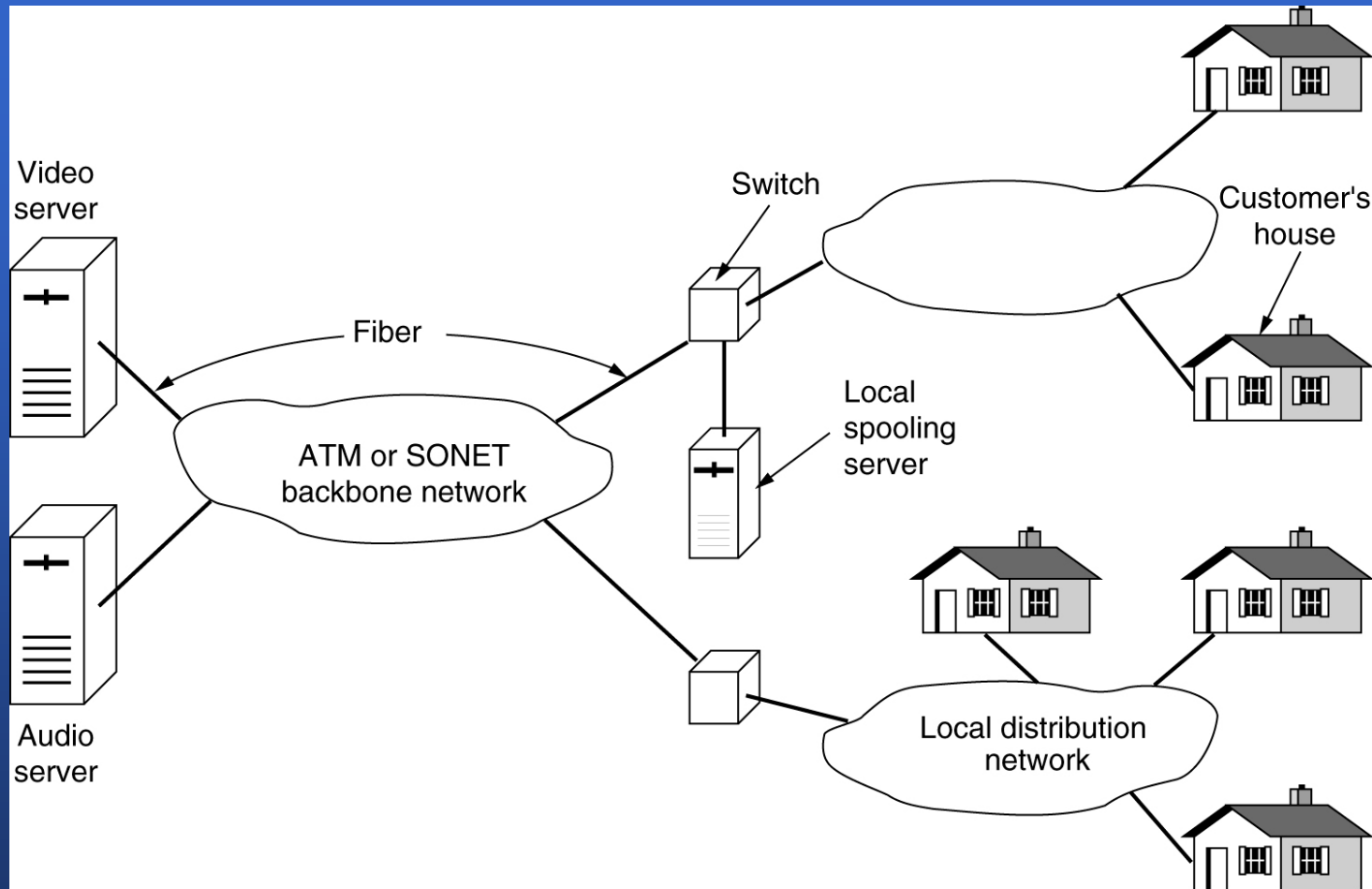
Synchronization of the audio and video streams in MPEG-1.

The MPEG Standard (2)



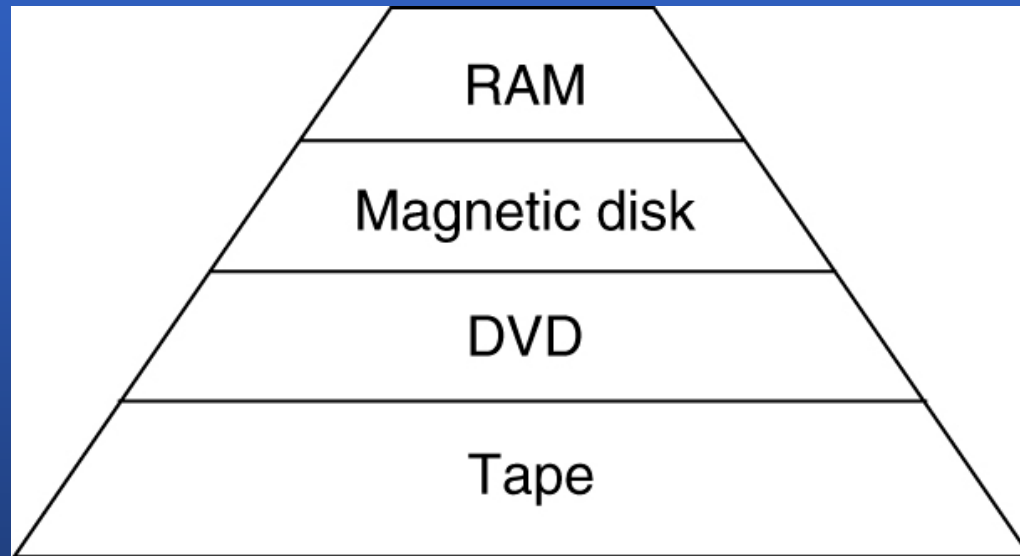
Three consecutive frames.

Video on Demand



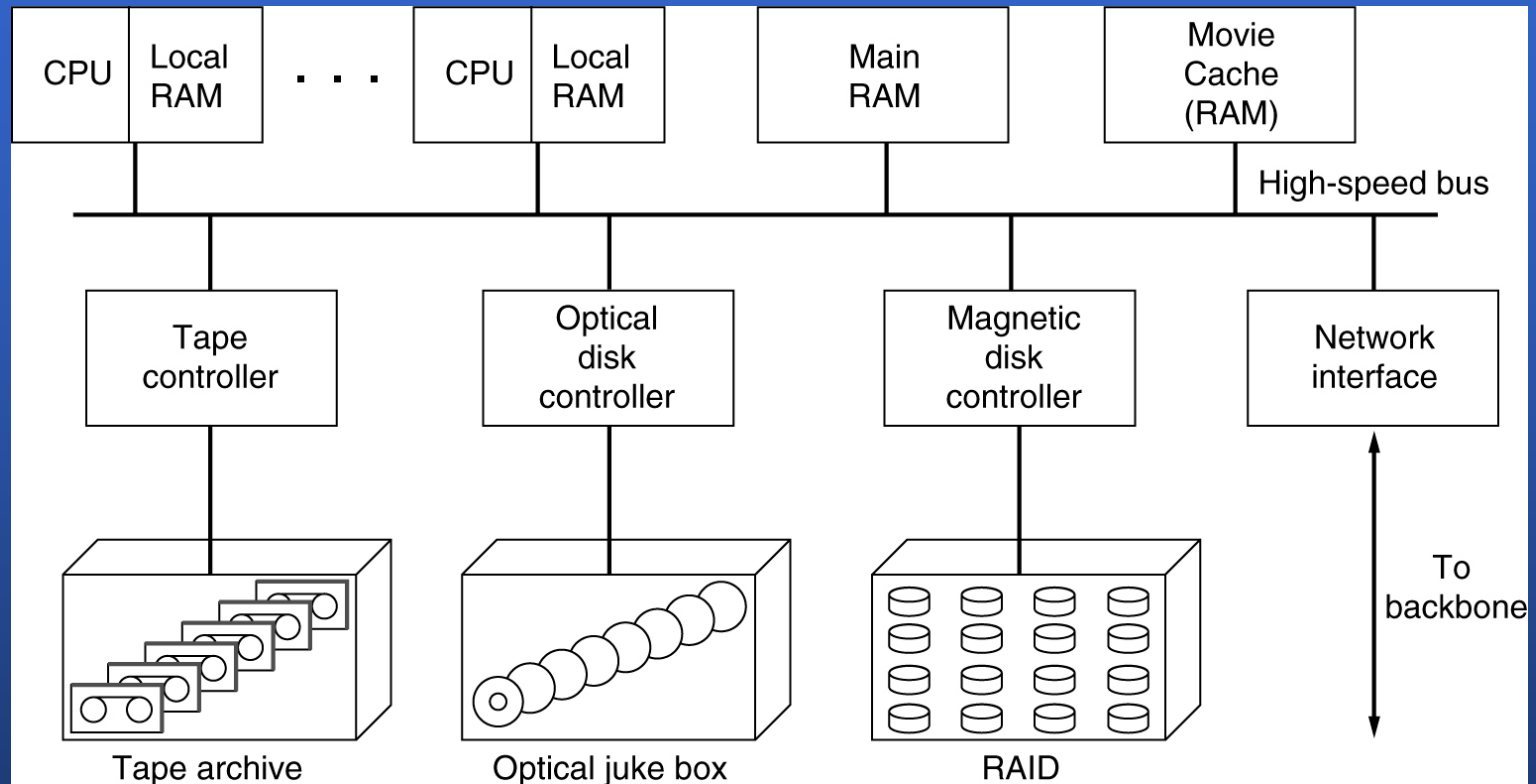
Overview of a video-on-demand system.

Video Servers



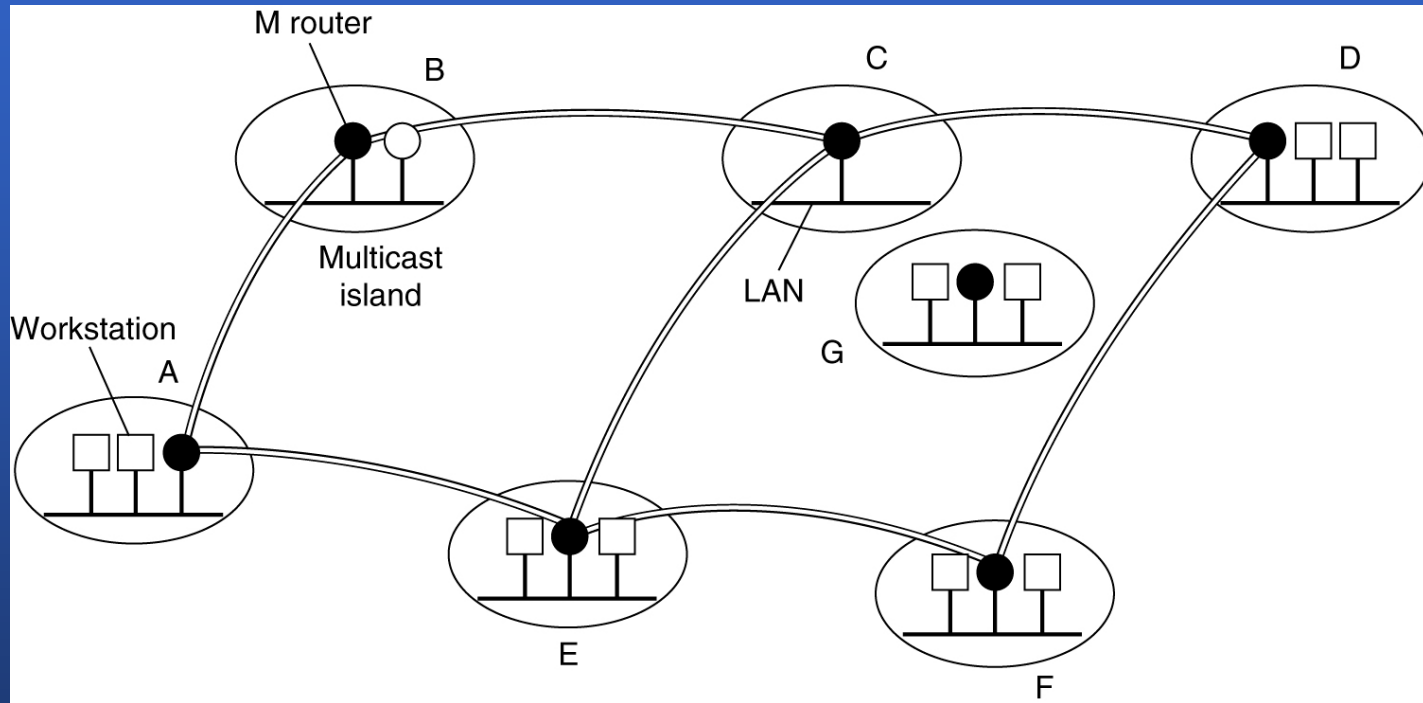
A video server storage hierarchy.

Video Servers (2)



The hardware architecture of a typical video server.

The MBone – The Multicast Backbone



MBone consists of multicast islands connected by tunnels.

7.5 SNMP—简单网络管理协议 (*)

- 一般，通过 ping、telnet 等访问路由器
- 互联网的扩大，使网络的管理越来越复杂
- SNMPv1 (RFC 1155)
- SNMPv2 (RFC 1441-1452)
- SNMPv3



7.5.1 SNMP模型

- SNMP模型由四部分组成：- [see fig 7-30]
 - 管理节点：主机、路由器、网桥、打印机。。。- 可运行SNMP管理进程，SNMP代理(agent)
 - 管理站：运行管理软件的计算机

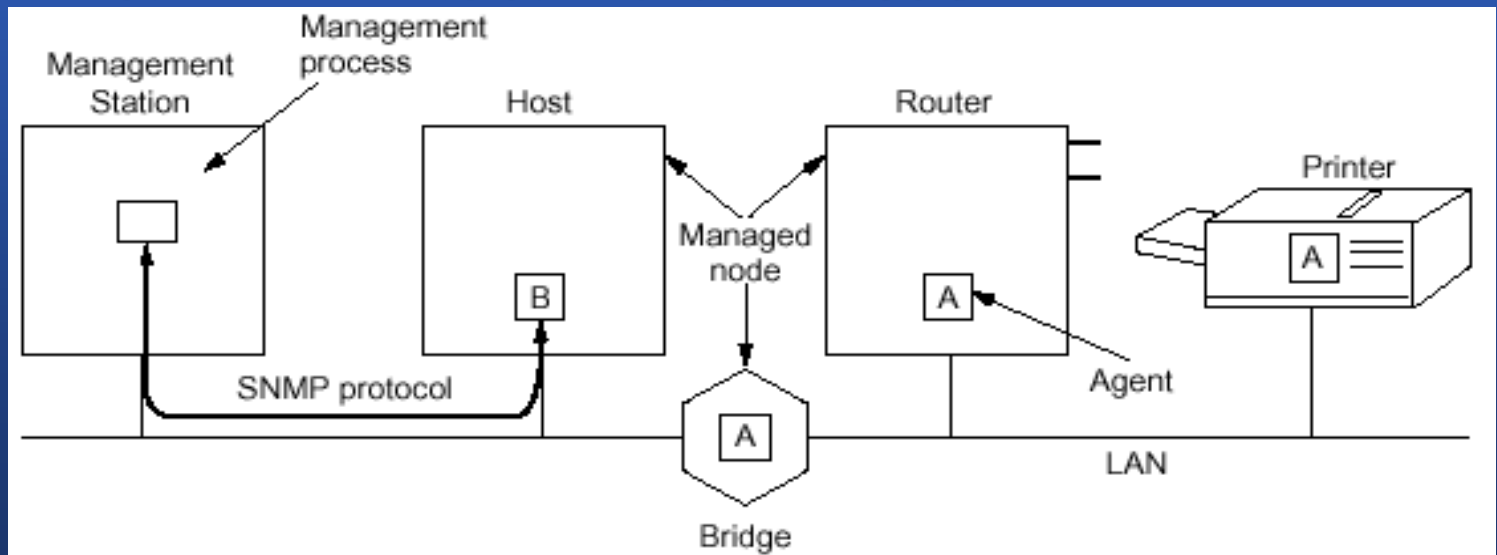
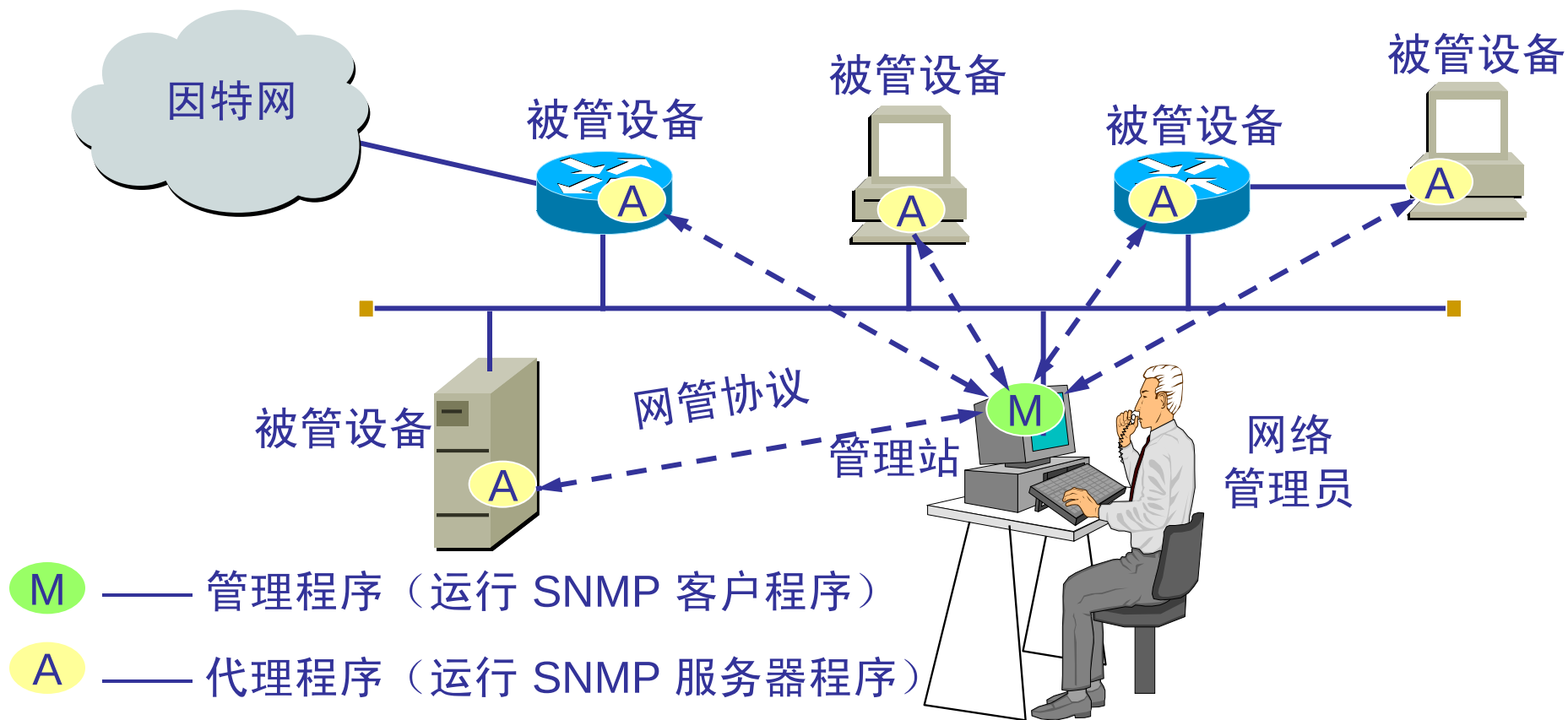


Fig. 7-30. Components of the SNMP management model.

网络管理的一般模型



7.5.1 SNMP模型

- SNMP模型

- 管理信息:

- 每个设备均由一个或多个变量描述其状态
- 变量也叫: 对象(object)
- 对象都放在管理信息库MIB中

- 管理协议: 实现管理站与代理的通信

- SNMP陷阱(trap)



7.5.2 ASN.1--抽象语法符号1

- SNMP的核心是由代理管理，由管理站读写的对象的集合
- SNMP使用抽象语法符号.1 (ASN.1)
- ASN.1基本数据类型-[see fig 7-31]

Primitive type	Meaning	Code
INTEGER	Arbitrary length integer	2
BIT STRING	A string of 0 or more bits	3
OCTET STRING	A string of 0 or more unsigned bytes	4
NULL	A place holder	5
OBJECT IDENTIFIER	An officially defined data type	6

Fig. 7-31. The ASN.1 primitive data types permitted in SNMP.

7.5.2 ASN.1--抽象语法符号1

- 部分ASN.1对象命名树-[see fig 7-32]

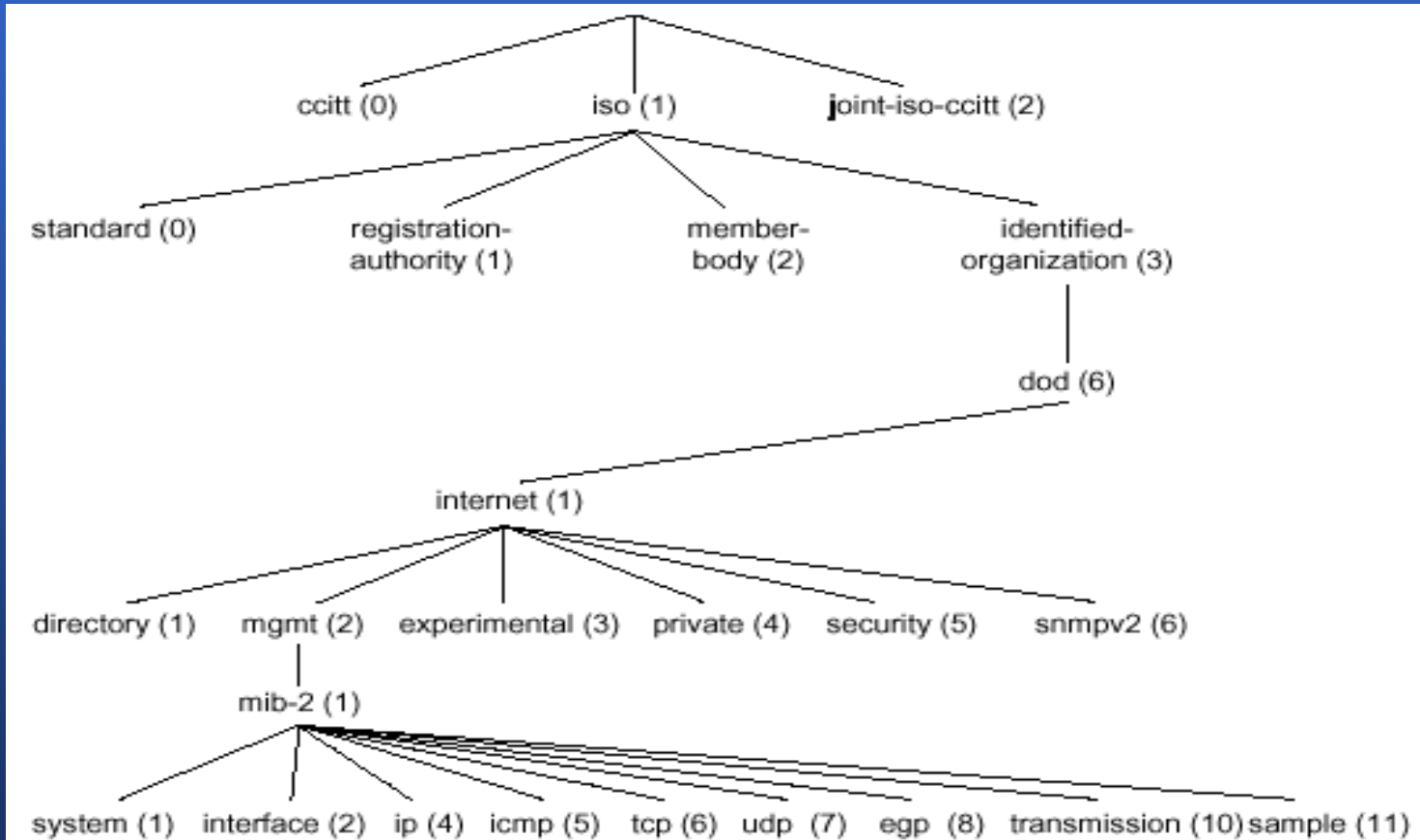


Fig. 7-32. Part of the ASN.1 object naming tree.

7.5.2 ASN.1--抽象语法符号1

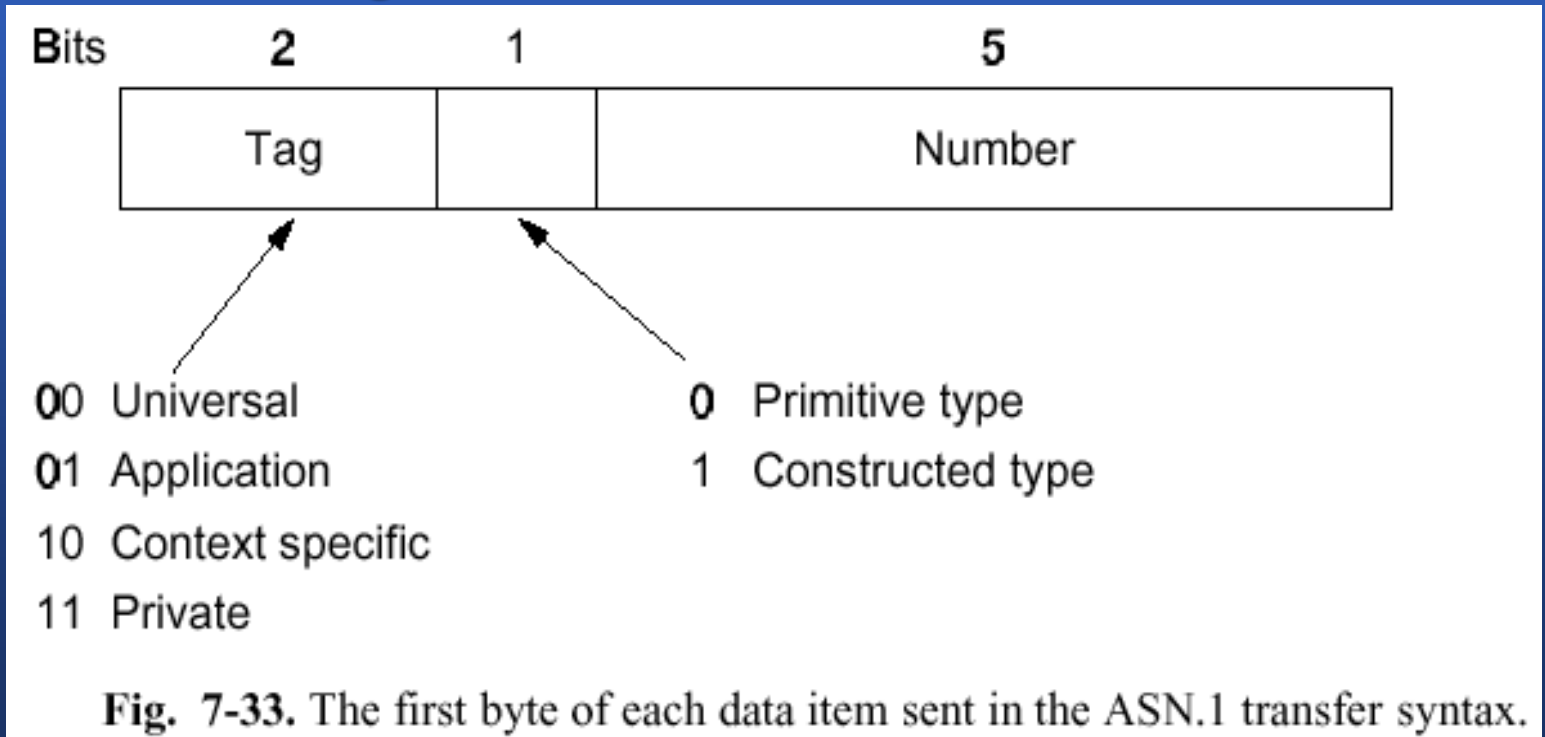
- ASN.1定义的由基本类型创建新类型的方法:
 - SEQUENCE
 - SEQUENCE OF
 - SET
 - SET OF
 - CHOICE
- ASN.1转换语法
 - 定义了ASN.1类型的值如何明确地转换为适于传输的字节序列
 - 使用的语法: 基本编码规则(BER)
 - BER的指导原则: 每个传输的值, 无论原语还是构造的, 最多由4个字段组成
 - 标识符
 - 数据字段的长度
 - 数据字段
 - 结束标志



7.5.2 ASN.1—抽象语法符号1

– ASN.1转换语法

- 由ASN.1转换语法发送的每个数据项的第一个字节—
[see fig 7-33]



7.5.2 ASN.1—抽象语法符号1

● 7.3.2 ASN.1—抽象语法符号1

– 一些示例值的ASN.1编码-[see fig 7-34]

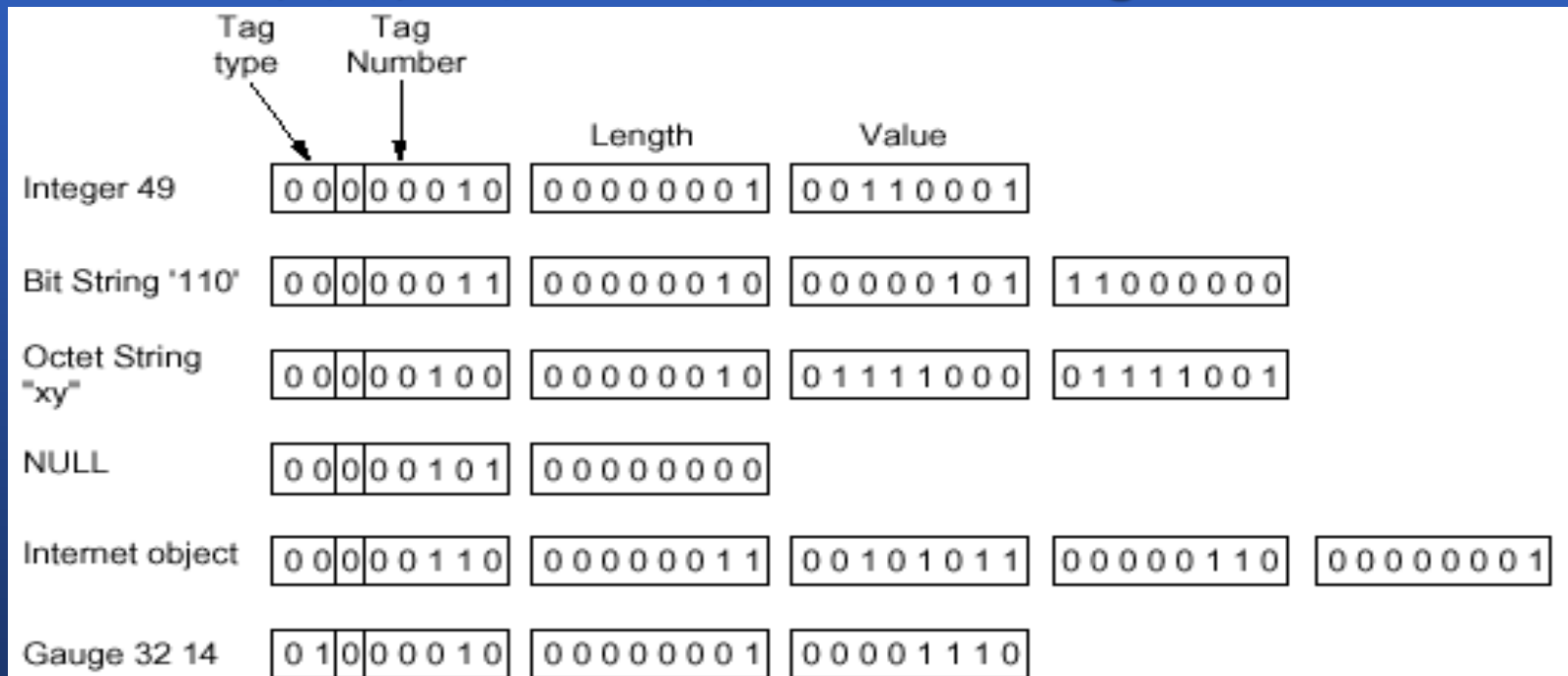


Fig. 7-34. ASN.1 encoding of some example values.

7.5.3 SMI—管理信息的结构

- SNMP变量被定义为单个对象
- 相关对象被集合为组 例如：IP对象组、TCP对象组
- 组被封装到模块中
- SNMP管理的变量的数据类型-[see fig 7-35]



Name	Type	Bytes	Meaning
INTEGER	Numeric	4	Integer (32 bits in current implementations)
Counter32	Numeric	4	Unsigned 32-bit counter that wraps
Gauge32	Numeric	4	Unsigned value that does not wrap
Integer32	Numeric	4	32 Bits, even on a 64-bit CPU
UInteger32	Numeric	4	Like Integer32, but unsigned
Counter64	Numeric	8	A 64-bit counter
TimeTicks	Numeric	4	In hundredths of a second since some epoch
BIT STRING	String	4	Bit map of 1 to 32 bits
OCTET STRING	String	≥ 0	Variable length byte string
Opaque	String	≥ 0	Obsolete; for backward compatibility only
OBJECT IDENTIFIER	String	> 0	A list of integers from Fig. 7-0
IpAddress	String	4	A dotted decimal Internet address
NsapAddress	String	< 22	An OSI NSAP address

Fig. 7-35. Data types used for SNMP monitored variables.

7.5.3 SMI—管理信息的结构

– SNMP变量的例子–[see fig 7-36]

lostPackets OBJECT TYPE

SYNTAX Counter32-- use a 32-bit counter

MAX-ACCESS read-only-- the management station may not change

STATUS current -- this variable is not obsolete (yet)

DESCRIPTION

"The number of packets lost since the last boot"

::= {experimental 20}

Fig. 7-36. An example SNMP variable.

7.5.4 MIB—管理对象库

- SNMP管理的对象集合定义在MIB库中，被分为10种
- MIB-II定义了175个对象
- MIB-II的对象组-[see fig 7-37]

Group	# Objects	Description
System	7	Name, location, and description of the equipment
Interfaces	23	Network interfaces and their measured traffic
AT	3	Address translation (deprecated)
IP	42	IP packet statistics
ICMP	26	Statistics about ICMP messages received
TCP	19	TCP algorithms, parameters, and statistics
UDP	6	UDP traffic statistics
EGP	20	Exterior gateway protocol traffic statistics
Transmission	0	Reserved for media-specific MIBs
SNMP	29	SNMP traffic statistics

Fig. 7-37. The object groups of the Internet MIB-II.

7.5.5 SNMP协议

- SNMP使用的一般方法：
 - 管理站向代理发一个请求，索要信息或命令其修改状态
 - 代理发回所要的信息，或证实已按要求修改了状态
 - 数据用ASN.1转换语法发送
 - 可能有各种错误的报告



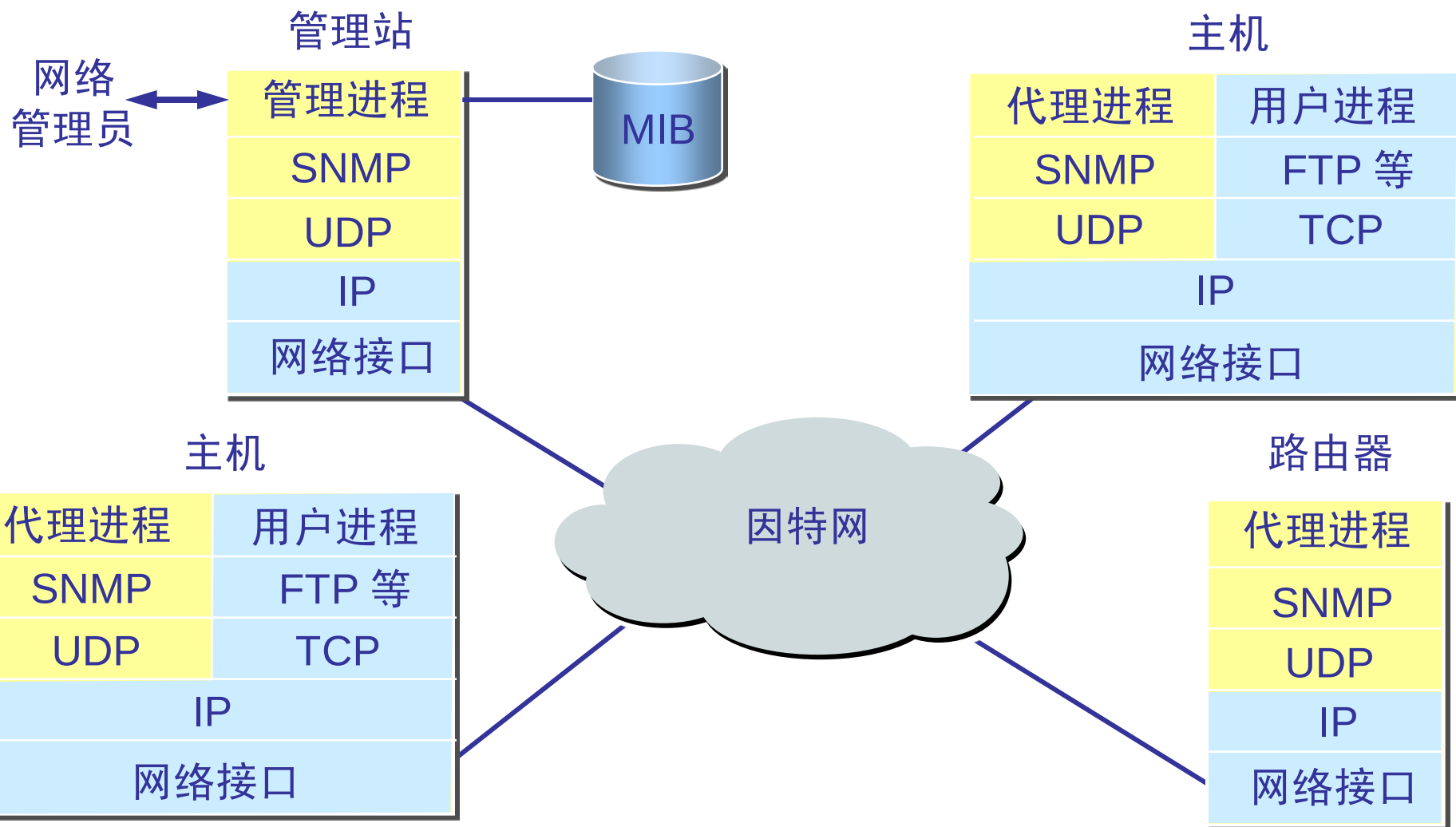
7.5.5 SNMP协议

- SNMP定义了可以发送的7种消息-[see fig 7-38]

Message	Description
Get-request	Requests the value of one or more variables
Get-next-request	Requests the variable following this one
Get-bulk-request	Fetches a large table
Set-request	Updates one or more variables
Inform-request	Manager-to-manager message describing local MIB
SnmpV2-trap	Agent-to-manager trap report

Fig. 7-38. SNMP message types.

SNMP 的典型配置



SNMP 使用的端口

- SNMP 使用无连接的 UDP，因此在网络上传送 SNMP 报文的开销较小。但 UDP 不保证可靠交付。
- 在运行代理程序的服务器端用通用端口 **161** 来接收 **get** 或 **set** 报文和发送响应报文（与通用端口通信的客户端使用临时端口）。
- 运行管理程序的客户端则使用通用端口 **162** 来接收来自各代理的 **trap** 报文。



SNMPv2 和 SNMPv3

SNMPv1 的主要缺点是：

- (1) 不能有效地传送大块的数据
- (2) 不能将网络管理的功能分散化
- (3) 安全性不够好



SNMPv2

- 1996 年发布 IETF 发布了 8 个 SNMPv2 文档[RFC 1901~1908]。但 SNMPv2 在安全方面的设计过分复杂，使得有些人不愿意接受它。
- SNMPv2 增加了 `get-bulk-request` 命令，可一次从路由器的路由表中读取许多行的信息。
- SNMPv2 的 `get` 命令允许返回部分的变量值，这就提高了效率，减少了网络上的通信量。
- SNMPv2 采用了分散化的管理方法。在一个网络中可以有多个顶级管理站，叫做管理服务器。



SNMPv2

- 增加了一个 **inform** 命令和一个管理进程到管理进程的 **MIB** (manager-to-manager MIB)。
- 使用这种 **inform** 命令可以使管理进程之间互相传送有关的事件信息而不需要经过请求。这样的信息则定义在管理进程到管理进程的 **MIB**中。



SNMPv3

- 1998 年 1 月 IETF 发表了 SNMPv3 的有关文档[RFC 2271-2275]。
- 仅隔 15 个月后就更新为[RFC 2571-2575]。
- SNMPv3 最大的改进就是安全特性。也就是说，只有被授权的人员才有资格执行网络管理的功能（如关闭某一条链路）和读取有关网络管理的信息（如读取一个配置文件的内容）。



● **7, 13, 27**

